

Higher risk, better filters - but which upgrade?

Nine new MT5 backtests + two locked models + one prior challenger | 2017-01-01 to 2026-09-09 | USD 100,000

T15 (0.65 ATR, 1.5% risk, range + time filters) is the preferred new balanced-growth candidate. It improves on the locked 0.25 ATR model in both CAGR and reported equity drawdown. The locked 0.65 ATR model remains the lower-drawdown choice.

Role	Model	CAGR %	Equity DD %	CAGR / Eq DD
Existing conservative	L65	13.40	13.17	1.018
Existing growth	L25	16.86	21.05	0.801
Preferred new balanced growth	T15	21.23	18.55	1.145
Conditional higher growth	T20	28.56	24.11	1.185

The filtered ATR family improves historical efficiency at each matching risk level. The 2.5% version leads the return/DD ratio ranking, but its 29.35% equity drawdown is too large to call conservative. Raising risk alone does not create a new edge.

These are development-sample results, not fresh out-of-sample validation. All nine are Gold ORB variants. They do not fill nine independent portfolio slots. The final section explicitly compares them with the locked systems and gives the selection decision.

Contents: source audit and model map; comparative performance; scaling and filter attribution; nine three-page model studies; stress tests and portfolio combinations; methodology; final locked-model comparison. Full daily/weekly/monthly/yearly ledgers are embedded as a ZIP attachment in the PDF.

Model map: actual settings, not filename guesses

All clocks below are broker time. Entry eligibility and scheduled position closure are different controls.

ID	Stop	Risk %	Entry eligibility	Range filter	Source filename
U15	0.65 D1 ATR	1.5	After 06:05 to 18:55	OFF	UF_65_15RISK.xlsx
U20	0.65 D1 ATR	2	After 06:05 to 18:55	OFF	UF_65_2RISK.xlsx
U25	0.65 D1 ATR	2.5	After 06:05 to 18:55	OFF	UF_65_25RISK.xlsx
R25	0.65 D1 ATR	2.5	After 06:05 to 18:55	0.00-0.60 ATR	UF_65_RANGE_25_RISK.xlsx
T15	0.65 D1 ATR	1.5	06:05-08:55	0.00-0.60 ATR	UF_65_RANGE_TIME_15RISK.xlsx
T20	0.65 D1 ATR	2	06:05-08:55	0.00-0.60 ATR	UF_65_RANGE_TIME_2RISK.xlsx
T25	0.65 D1 ATR	2.5	06:05-08:55	0.00-0.60 ATR	UF_65_RANGE_TIME_25RISK.xlsx
P20	1% of entry price	2	After 06:05 to 18:55	OFF	1PERCENTSL_2RISK.xlsx
PT20	1% of entry price	2	06:05-08:55	0.00-0.60 ATR	1PERCENTSL_WITHRANGETIME_2RISK.xlsx
L65	0.65 D1 ATR	1	After 06:05 to 18:55	OFF	UF_65_1_RISK.xlsx
L25	0.25 D1 ATR	0.6	After 06:05 to 18:55	OFF	UF_25_06_RISK.xlsx
P10	1% of entry price	1	After 06:05 to 18:55	OFF	UF_25_PRICE_SL1PERCENT.xlsx

Time-filtered rows use start 06:05 inclusive and end 08:55 exclusive. They are NOT 07:00-to-EOD or 07:30-to-09:00 tests. When the entry filter is OFF, retained 07:00 input values are inactive. All positions can continue to the 18:55 close unless stopped.

Audit: twelve reports reconciled

Nine new attachments are assessed against the original locked C/D choices and the prior price-stop challenger.

Check	Result
Cash and trades	All entry/exit pairs reconstructed; unique deal IDs; no unmatched positions. Trade counts, net/gross P&L, every balance and relative balance DD reconcile with MT5.
Comparability	Same configured dates, USD 100,000, 1:100 leverage, 99% history quality, 227,831 bars and 346,317,299 ticks. Same symbol and source company.
Version bridge	New reports use ORB_V5_ALLSL; L65 and L25 use OBR_V4. U15 and L65 have all 2,479 entries, exits, directions, fill prices and stops identical. This supports a clean historical risk-sizing comparison for ATR 0.65.
Stop selection	Stop_Type=0 selects ATR for U/R/T. Stop_Type=1 selects price percentage for P/PT; the stored ATR multiplier is inactive in these runs.
Holding and costs	At most one completed trade per date; no overnight trades. Commission and swap recorded as zero. Spread may be reflected in fill prices.
What is not supplied	No intratrade equity time series, native tester journal, broker volume/margin specification, or binary equivalence proof. History quality does not identify real-tick versus generated-tick mode.

All analysis uses the relative equity DD field, not the percentage beside maximum currency DD. Independently reconstructed charts show closed balance. No synthetic intratrade equity curve is presented.

Performance dashboard

Net profit is in USD; CAGR uses the complete configured calendar period, including inactive weekdays.

ID	Trades	Net profit \$	CAGR %	PF	Win %	Mean R	Bal DD %	Eq DD %
U15	2,479	494,501	20.20	1.214	50.50	0.0524	18.33	19.27
U20	2,479	904,285	26.89	1.220	50.50	0.0525	24.00	25.16
U25	2,479	1,530,910	33.40	1.226	50.50	0.0525	29.34	30.80
R25	2,378	1,609,300	34.05	1.281	50.84	0.0550	29.89	31.28
T15	1,853	545,524	21.23	1.304	51.43	0.0717	16.93	18.55
T20	1,853	1,040,069	28.56	1.316	51.43	0.0718	22.21	24.11
T25	1,853	1,857,132	35.94	1.325	51.43	0.0718	27.27	29.35
P20	2,479	919,145	27.08	1.263	50.79	0.0518	23.87	25.86
PT20	1,853	858,015	26.27	1.336	51.59	0.0660	23.99	25.25
L65	2,479	238,117	13.40	1.206	50.50	0.0521	12.50	13.17
L25	2,479	352,483	16.86	1.220	37.64	0.1097	20.66	21.05
P10	2,479	237,127	13.37	1.240	50.79	0.0515	12.46	13.80

Risk-adjusted performance dashboard

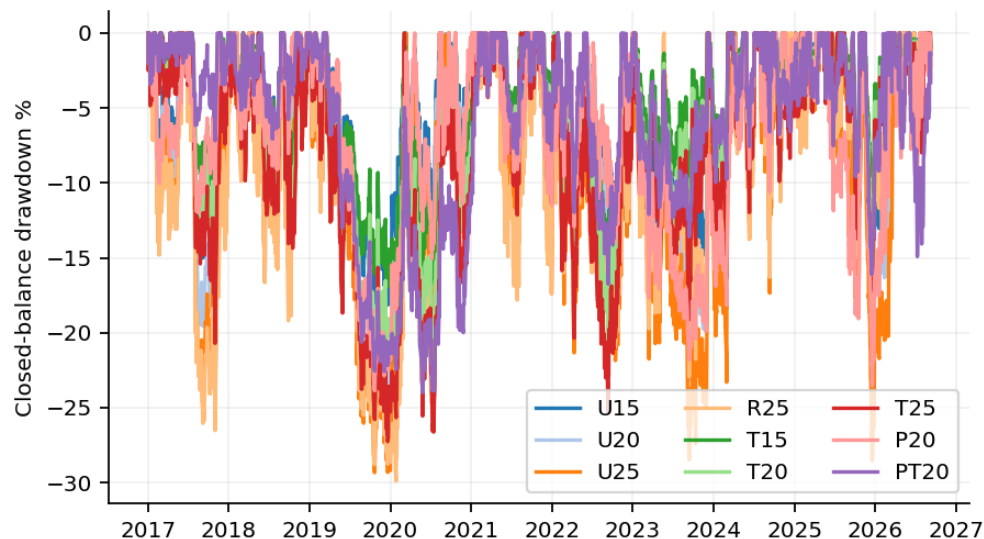
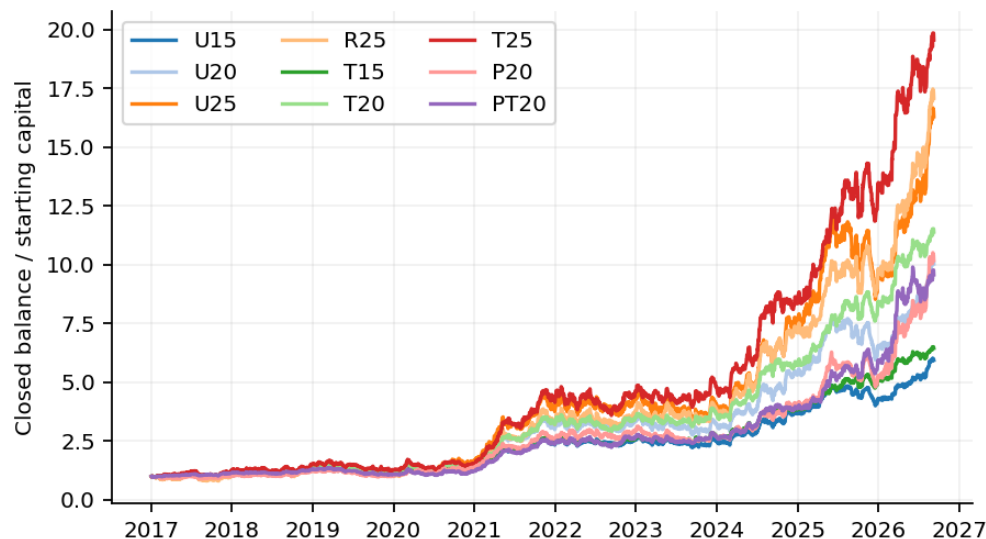
CAGR/drawdown ratios and daily-return ratios answer different questions.

ID	Sharpe	Sortino	Volatility %	CAGR / Bal DD	CAGR / Eq DD
U15	1.063	1.769	18.29	1.102	1.048
U20	1.063	1.770	24.42	1.121	1.069
U25	1.063	1.769	30.54	1.139	1.085
R25	1.107	1.841	29.48	1.139	1.089
T15	1.244	2.114	15.97	1.254	1.145
T20	1.244	2.115	21.32	1.286	1.185
T25	1.245	2.115	26.67	1.318	1.224
P20	1.152	1.932	22.23	1.135	1.047
PT20	1.258	2.150	19.40	1.095	1.041
L65	1.059	1.762	12.16	1.072	1.018
L25	1.036	2.277	15.72	0.816	0.801
P10	1.151	1.929	11.06	1.072	0.969

Sharpe compares average daily return with daily variability. CAGR/DD compares compounded annual growth with the worst observed loss from a prior peak. A higher Sharpe can coexist with a lower CAGR/DD, as the price-stop filtered candidate demonstrates. Neither replaces an explicit absolute drawdown limit.

Growth and drawdown across the nine new runs

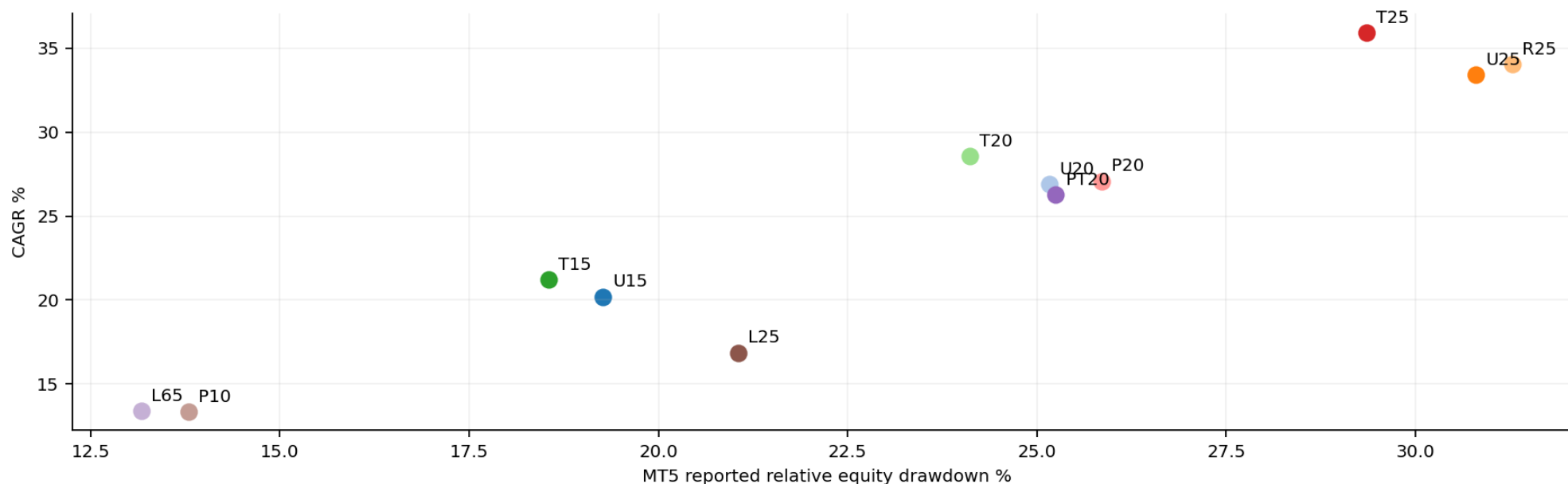
Every curve starts from the same capital. Compounded dollars exaggerate visual gaps at higher risk.



U15/U20/U25 share the same signals. T15/T20/T25 also share signals. The large separation in final wealth within either family mainly reflects sizing and compounding; their Sharpe ratios are almost unchanged within the family.

Return versus reported equity drawdown

The selection objective is a useful return with a tolerable loss path, not the largest ratio at any risk.



T15 is above and left of L25: higher CAGR and lower equity DD. It is above and right of L65: higher CAGR but more drawdown. T20 and T25 offer more growth only by accepting progressively deeper losses. A ratio near 1.2 does not make a 29% drawdown small.

Does raising risk improve the strategy?

Actual reruns, not linearly multiplied profit totals

Family	Risk %	CAGR %	Eq DD %	CAGR / Eq DD	Sharpe	Mean budget R
L65	1	13.40	13.17	1.018	1.059	0.0521
U15	1.5	20.20	19.27	1.048	1.063	0.0524
U20	2	26.89	25.16	1.069	1.063	0.0525
U25	2.5	33.40	30.80	1.085	1.063	0.0525
T15	1.5	21.23	18.55	1.145	1.244	0.0717
T20	2	28.56	24.11	1.185	1.244	0.0718
T25	2.5	35.94	29.35	1.224	1.245	0.0718

Unfiltered sizing increases from 1% to 2.5% move CAGR from 13.40% to 33.40%, but equity DD from 13.17% to 30.80%. The ratio rises only from 1.018 to 1.085. The normalized edge and Sharpe remain essentially unchanged.

The filtered family has Sharpe near 1.244 versus approximately 1.063 for the higher-risk unfiltered family. This is evidence of a historical selection benefit beyond just leverage. Within the filtered family, however, the higher return/DD ratio at higher risk still does not prove a better signal.

Risk budgets and realized losses differ. Gap/slippage can exceed the nominal stop risk; lot rounding makes average initial exposure slightly smaller. The report includes observed risk and worst-trade checks for each run. No return scaling is treated as an exact new MT5 backtest.

Do the filters earn their complexity?

Matched entry dates separate removed opportunities from risk compounding.

Parent -> child	Removed trades	Removed net R	Profit retained %	Eq DD change (pp)
U15_T15	626	-2.823	110.32	-0.72
U20_T20	626	-2.830	115.02	-1.05
U25_T25	626	-2.808	121.31	-1.45
U25_R25	101	-0.585	105.12	0.48
R25_T25	525	-2.223	115.40	-1.93
P20_PT20	626	6.012	93.35	-0.61

Negative removed R means the excluded parent trades lost in aggregate after normalizing to their risk budget. Dollar profit retention can exceed 100% because the filtered path compounds differently. Removed R is diagnostic, not an additive decomposition of compounded dollar profits.

For ATR 0.65, both filters improve CAGR and reduce equity DD at all three risk settings. For a 1%-of-price stop at 2% risk, both filters reduce CAGR and only slightly reduce equity DD. The same entry filter can help one exit model and hurt another: losing and winning exits change with stop distance.

Only U25 -> R25 isolates the range cap. Only R25 -> T25 isolates the time cutoff after that range cap. U15 -> T15 and U20 -> T20 change both filters together; they cannot identify each filter separately. A time-only ATR 0.65 run is not supplied in this batch.

Which years supplied the filter benefit?

Removed trades in budget R, based on the parent ledger; not dollar attribution.

Year	U25_R25	R25_T25	U25_T25	P20_PT20
2017	1.369	-7.609	-6.239	-5.440
2018	1.641	3.811	5.445	2.834
2019	-0.536	-3.520	-4.063	-0.755
2020	3.204	6.422	9.628	10.504
2021	1.197	-7.659	-6.464	-1.911
2022	-0.350	2.880	2.538	1.463
2023	-3.874	-0.596	-4.472	-4.704
2024	0.215	2.678	2.897	1.736
2025	-2.502	-6.012	-8.514	-7.219
2026	-0.949	7.382	6.436	9.503

A negative cell favours exclusion in that year; a positive cell shows forgone edge. The time cutoff trades away late-session opportunities that can be valuable in some years. It must not be described as universally removing bad trades.

The 06:05-08:55 entry window still keeps the 06:05-07:00 period. These results therefore do not answer the older question of excluding 06:00-07:00 or testing 07:00-to-EOD. Do not transfer this conclusion to those different settings.

Year-by-year return comparison

Returns are percentages. 2026 is partial through 8 September and is not annualized here.

Year	U15	U20	U25	R25	T15	T20	T25	P20	PT20	L65	L25	P10
2017	5.4	6.7	7.6	4.1	16.2	21.5	26.9	3.6	15.9	3.9	2.3	2.0
2018	20.8	27.6	34.3	29.1	11.9	15.4	18.8	17.8	11.8	13.8	25.1	9.0
2019	-15.0	-20.1	-25.1	-24.0	-9.4	-12.8	-16.2	-17.9	-16.3	-10.0	-12.2	-9.0
2020	30.6	41.8	53.0	42.1	13.7	18.0	22.0	37.8	12.6	19.9	15.2	18.3
2021	66.8	96.4	130.2	123.6	84.4	124.8	173.0	87.1	95.1	41.2	33.4	37.6
2022	10.4	13.1	15.3	16.6	6.7	8.3	9.5	15.4	12.7	7.2	28.0	8.2
2023	-3.5	-5.5	-7.8	1.7	3.6	4.1	4.3	-9.1	0.4	-2.0	-8.7	-4.2
2024	46.0	64.4	84.4	84.5	40.5	56.5	73.9	48.3	44.2	29.1	29.7	22.2
2025	17.7	23.5	29.1	38.4	34.3	47.5	61.5	31.6	53.2	11.8	26.9	15.6
2026	37.8	52.8	69.1	73.8	25.4	35.0	45.0	92.7	61.1	23.7	36.3	39.9

2019 is a losing year for every new model and both locked models. The filtered ATR group reduces 2019 losses and turns 2023 positive, but gives up some strength in years such as 2020 and 2026. This is an altered distribution of returns, not immunity to unfavourable periods.

Segment	U15	U20	U25	R25	T15	T20	T25	P20	PT20
2017-2019	8.2	8.8	8.2	2.2	17.7	22.3	26.3	0.3	8.4
2020-2022	140.4	214.8	306.1	270.5	123.7	187.4	264.9	197.5	147.5
2023	-3.5	-5.5	-7.8	1.7	3.6	4.1	4.3	-9.1	0.4
2024-2026	136.9	210.4	302.6	343.6	136.6	211.7	307.3	276.0	255.7

The lower table compounds the full segment. These chronological slices are diagnostic only: the same history has already influenced model selection, so none is newly untouched validation.

Consistency, concentration and recovery

Positive months, losing periods and dependence on the strongest year

ID	Positive months	Worst month %	Worst week %	Longest DD days	Top 10 R share %	CAGR if 2021 flat %
U15	65/117	-8.47	-6.53	433	25.98	14.02
U20	65/117	-11.20	-8.63	438	25.96	18.35
U25	64/117	-13.87	-10.70	440	25.97	22.40
R25	66/117	-13.88	-10.70	553	25.31	23.37
T15	78/117	-7.93	-5.85	351	24.23	13.81
T20	76/117	-10.48	-7.76	351	24.22	18.25
T25	76/117	-12.99	-9.64	353	24.23	22.55
P20	66/117	-9.26	-7.76	496	27.30	19.12
PT20	72/117	-7.73	-7.30	689	28.31	17.85
L65	65/117	-5.69	-4.38	432	26.04	9.43
L25	64/117	-6.46	-2.99	552	30.41	13.44
P10	67/117	-4.69	-3.92	455	27.26	9.69

Longest drawdown is calendar time from a closed-balance peak to recovery, or to the test end if unrecovered. It is not the duration of the deepest intratrade equity drawdown. T15/T20 recover more quickly than L25 in this sample, while PT20 remains below a prior balance high much longer.

The 2021-flat scenario replaces that year's daily returns with zero and compounds all other recorded returns over the unchanged test horizon. It measures dependence on a strong year; it is not a deployable rule or a counterfactual execution backtest.

At the same historical drawdown, does the edge improve?

Return-path replay at L65's 12.50% BALANCE drawdown; equity DD is unavailable for synthetic paths.

Source	Approx. risk %	Replay CAGR %	Balance DD %	CAGR / Bal DD
L65 actual	1	13.40	12.50	1.072
T15	1.092	15.34	12.50	1.227
U15	1.002	13.51	12.50	1.080
L25	0.354	9.97	12.50	0.798
P20	0.997	13.41	12.50	1.073
PT20	1.010	13.04	12.50	1.043

The filtered T15 path gives approximately 15.34% CAGR at the same historical balance DD as L65, versus 13.40% for L65. That supports a historical filter benefit at comparable risk. The equivalent replay risk of roughly 1.09% is a mathematical match, not a recommended finely optimized parameter.

A simpler 1% replay of T15 gives 14.03% CAGR and 11.48% balance DD. It has not been rerun in MT5. If a low-drawdown replacement for L65 is desired, a real 1% filtered run is the direct next comparison.

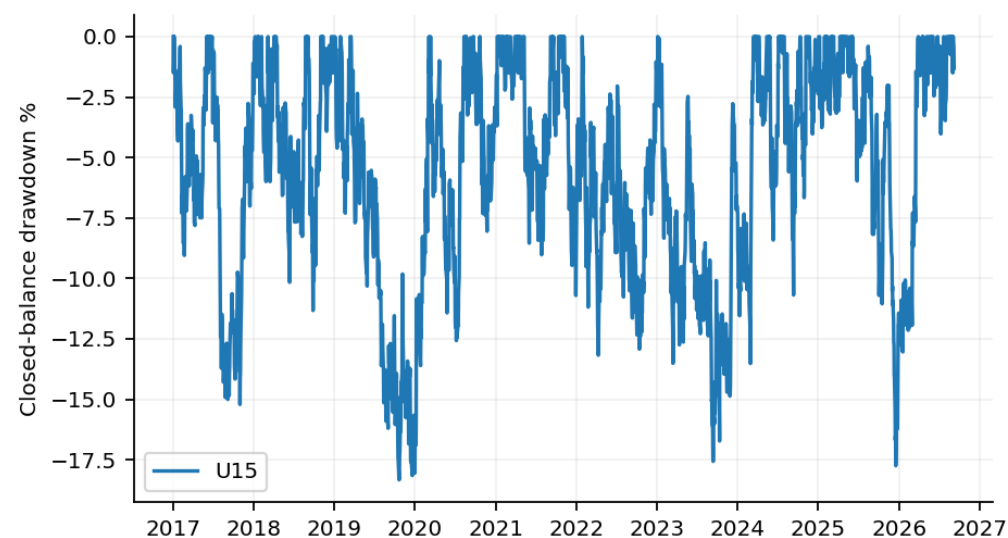
Replay scales recorded daily returns before compounding. It cannot reproduce new lot rounding, margin constraints, trade rejection, broker volume limits, intratrade equity or liquidity effects. Exact matched-DD sizing is intentionally not promoted as a production setting.

U15 | ATR 0.65 | unfiltered | risk 1.5%

Individual study 1/3 - performance, actual closed-balance curve and decision

Improves growth over L65 through higher sizing. T15 has higher CAGR and lower equity drawdown at the same risk, so U15 is not the preferred new candidate.

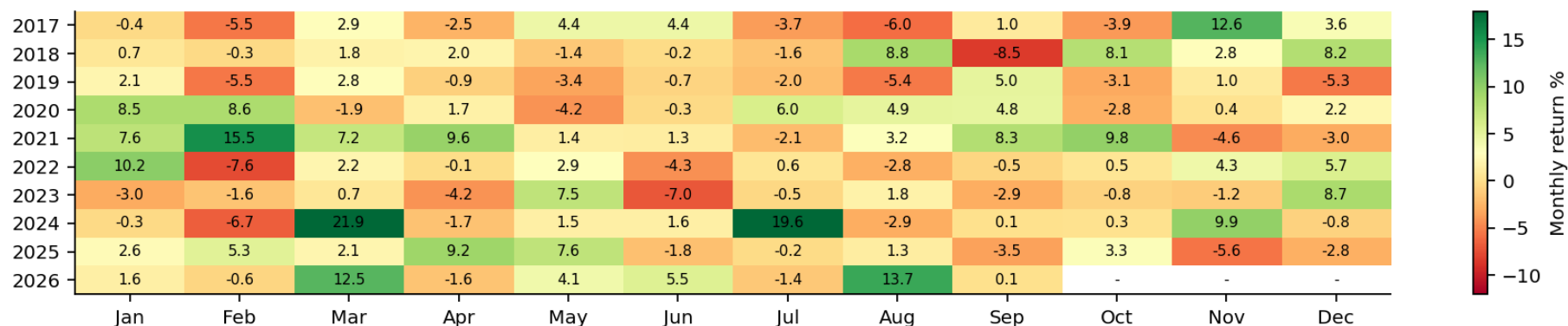
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
494,501.16	594,501.16	20.20	2,479	1.214	50.50
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
18.33	19.27	1.048	1.063	1.769	18.29



Average budget expectancy: 0.0524R/trade. Average winner: 0.647R; average loser: -0.554R. Trades/month across the full horizon: 21.3. Longest win/loss streaks: 14/11. Average holding time: 10.52 hours.

U15 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

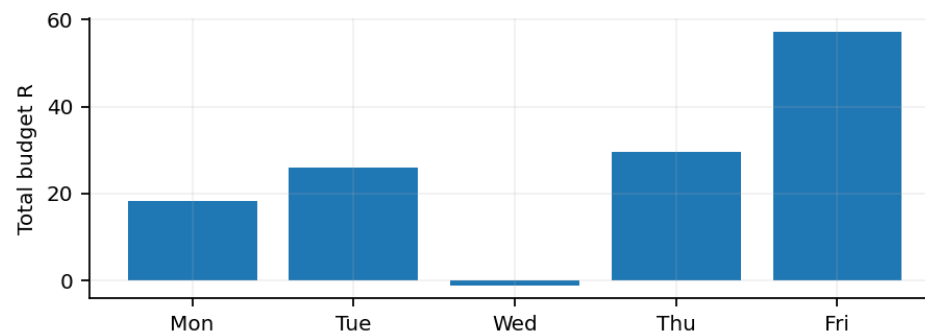
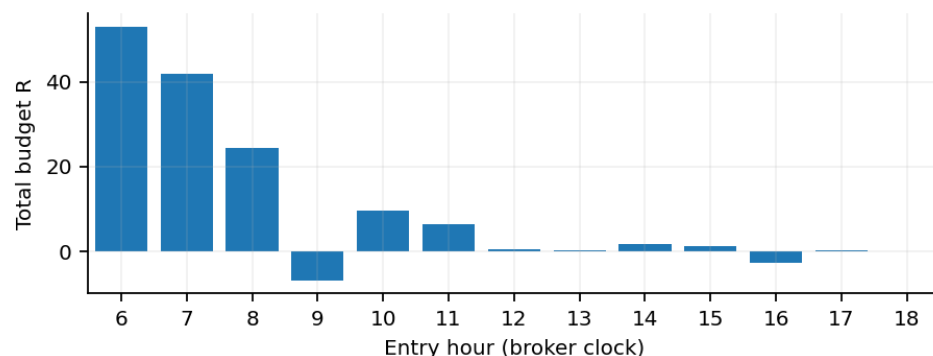


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	255	5,446	5.45	15.21	1.053	0.0175
2018	257	21,905	20.77	11.33	1.185	0.0537
2019	256	-19,166	-15.05	18.33	0.859	-0.0388
2020	257	33,076	30.57	12.57	1.244	0.0741
2021	257	94,387	66.82	10.71	1.475	0.1377
2022	257	24,411	10.36	13.18	1.081	0.0308
2023	256	-9,106	-3.50	17.57	0.969	-0.0043
2024	258	115,520	46.03	10.69	1.382	0.1025
2025	254	64,979	17.73	17.75	1.139	0.0472
2026	172	163,049	37.79	4.02	1.625	0.1279

Positive months: 65/117; positive weeks: 291/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (21.86%); worst: 2018-09 (-8.47%). Best/worst weeks: 2023-12-04/2023-12-10 (10.42%) / 2018-09-10/2018-09-16 (-6.53%).

U15 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-10-22	2020-03-05	18.33	353
2025-06-05	2025-12-18	2026-03-30	17.75	298
2023-01-06	2023-09-13	2024-03-14	17.57	433

Side	Trades	Net profit \$	PF	Mean R
buy	1289	392,707	1.365	0.0772
sell	1190	101,794	1.082	0.0256

Actual initial SL exposure averaged 1.496% (range 1.462%-1.500%); worst closed trade -1.57%. Empirical daily VaR95/ES95: 1.50%/1.50%. 20-day block bootstrap balance-DD median / 95th percentile: 25.44% / 38.09%. These resampling levels are conditional stress estimates.

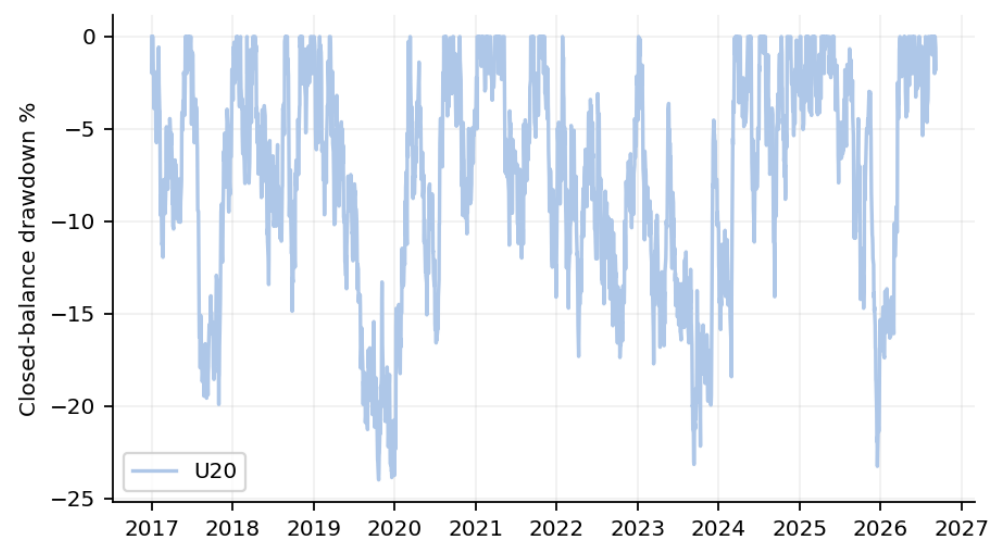
Rolling 3/6/12-month proxy worst returns: -14.44% / -14.53% / -15.51%; positive 12-month windows: 83.82%. Top 10 winners contribute 32.92% of net dollars and 25.98% of total R. With those top 10 R trades replaced by zero, replay CAGR is 14.23%.

U20 | ATR 0.65 | unfiltered | risk 2%

Individual study 1/3 - performance, actual closed-balance curve and decision

Higher exposure to the same unfiltered edge. T20 beats it on CAGR, equity drawdown and Sharpe; no independent portfolio role is demonstrated.

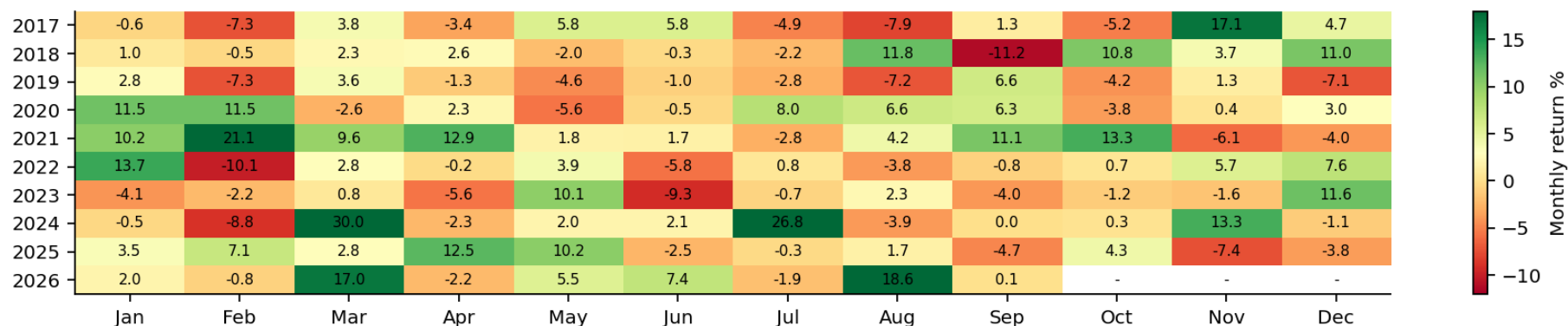
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
904,285.06	1,004,285.06	26.89	2,479	1.220	50.50
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
24.00	25.16	1.069	1.063	1.770	24.42



Average budget expectancy: 0.0525R/trade. Average winner: 0.648R; average loser: -0.555R. Trades/month across the full horizon: 21.3. Longest win/loss streaks: 14/11. Average holding time: 10.52 hours.

U20 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

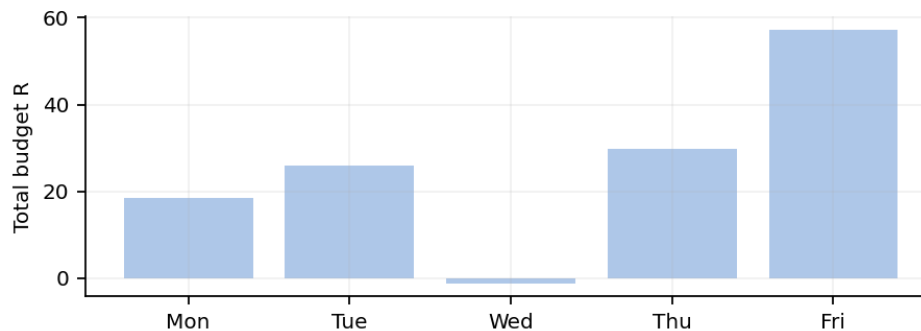
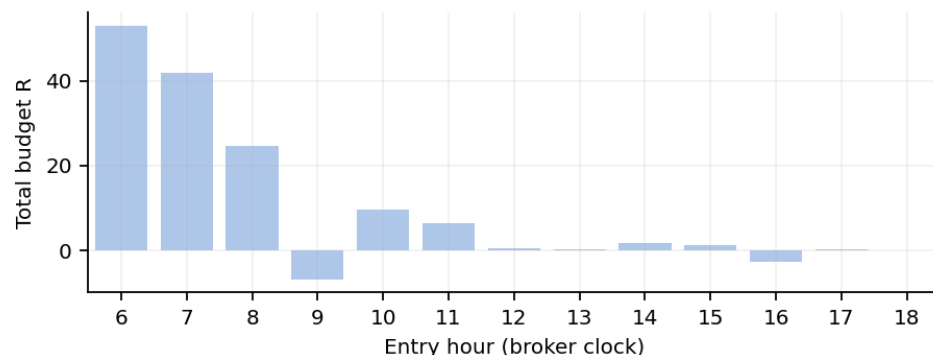


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	255	6,717	6.72	19.91	1.050	0.0176
2018	257	29,467	27.61	14.87	1.182	0.0537
2019	256	-27,403	-20.12	24.00	0.855	-0.0390
2020	257	45,463	41.79	16.58	1.236	0.0745
2021	257	148,621	96.35	14.10	1.446	0.1378
2022	257	39,575	13.07	17.37	1.075	0.0308
2023	256	-18,854	-5.51	23.16	0.962	-0.0045
2024	258	208,514	64.44	14.08	1.375	0.1026
2025	254	125,139	23.52	23.26	1.130	0.0474
2026	172	347,044	52.80	5.35	1.626	0.1281

Positive months: 65/117; positive weeks: 291/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (30.01%); worst: 2018-09 (-11.20%). Best/worst weeks: 2023-12-04/2023-12-10 (14.02%) / 2018-09-10/2018-09-16 (-8.63%).

U20 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-10-22	2020-03-12	24.00	360
2025-06-05	2025-12-18	2026-03-30	23.26	298
2023-01-06	2023-09-13	2024-03-19	23.16	438

Side	Trades	Net profit \$	PF	Mean R
buy	1289	730,699	1.386	0.0773
sell	1190	173,586	1.078	0.0257

Actual initial SL exposure averaged 1.996% (range 1.971%-2.000%); worst closed trade -2.10%. Empirical daily VaR95/ES95: 2.00%/2.00%. 20-day block bootstrap balance-DD median / 95th percentile: 32.81% / 47.90%. These resampling levels are conditional stress estimates.

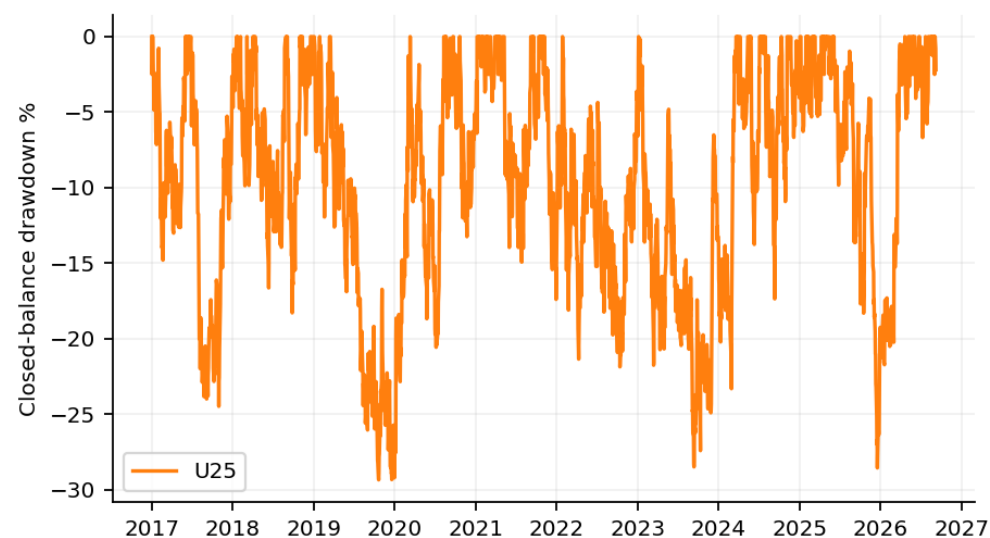
Rolling 3/6/12-month proxy worst returns: -18.97% / -19.18% / -20.68%; positive 12-month windows: 82.29%. Top 10 winners contribute 36.44% of net dollars and 25.96% of total R. With those top 10 R trades replaced by zero, replay CAGR is 18.61%.

U25 | ATR 0.65 | unfiltered | risk 2.5%

Individual study 1/3 - performance, actual closed-balance curve and decision

Large compounded returns, but 30.80% equity drawdown. The ratio improvement from sizing alone is small, and T25 has a better historical trade-off.

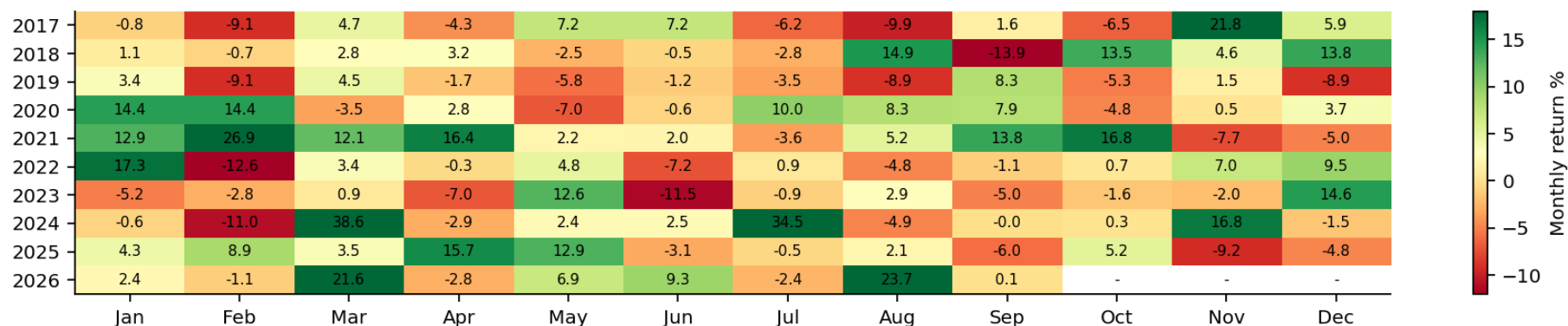
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
1,530,910.36	1,630,910.36	33.40	2,479	1.226	50.50
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
29.34	30.80	1.085	1.063	1.769	30.54



Average budget expectancy: 0.0525R/trade. Average winner: 0.648R; average loser: -0.555R. Trades/month across the full horizon: 21.3. Longest win/loss streaks: 14/11. Average holding time: 10.52 hours.

U25 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

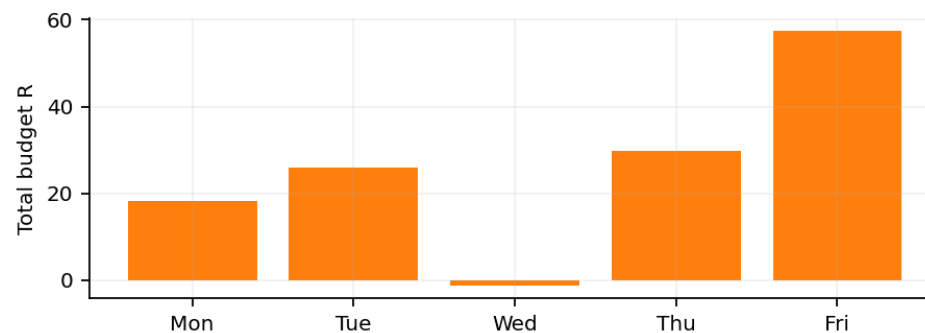
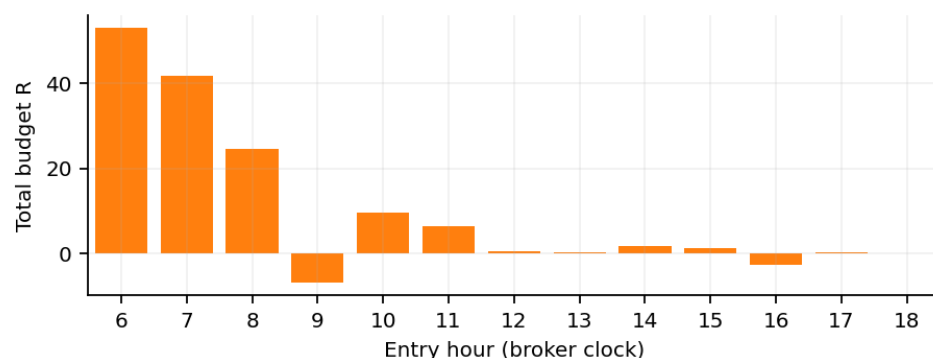


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	255	7,562	7.56	24.47	1.045	0.0175
2018	257	36,858	34.27	18.28	1.179	0.0537
2019	256	-36,260	-25.11	29.34	0.852	-0.0390
2020	257	57,305	52.98	20.56	1.226	0.0743
2021	257	215,469	130.22	17.39	1.418	0.1380
2022	257	58,296	15.30	21.85	1.070	0.0308
2023	256	-34,094	-7.76	28.48	0.956	-0.0044
2024	258	342,114	84.44	17.35	1.368	0.1027
2025	254	217,093	29.05	28.54	1.122	0.0474
2026	172	666,566	69.12	6.68	1.627	0.1283

Positive months: 64/117; positive weeks: 290/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (38.64%); worst: 2018-09 (-13.87%). Best/worst weeks: 2023-12-04/2023-12-10 (17.69%) / 2018-09-10/2018-09-16 (-10.70%).

U25 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-10-22	2020-03-12	29.34	360
2025-06-05	2025-12-18	2026-04-21	28.54	320
2023-01-06	2023-09-13	2024-03-21	28.48	440

Side	Trades	Net profit \$	PF	Mean R
buy	1289	1,259,650	1.408	0.0773
sell	1190	271,261	1.073	0.0257

Actual initial SL exposure averaged 2.497% (range 2.477%-2.500%); worst closed trade -2.62%. Empirical daily VaR95/ES95: 2.50%/2.51%. 20-day block bootstrap balance-DD median / 95th percentile: 39.67% / 56.50%. These resampling levels are conditional stress estimates.

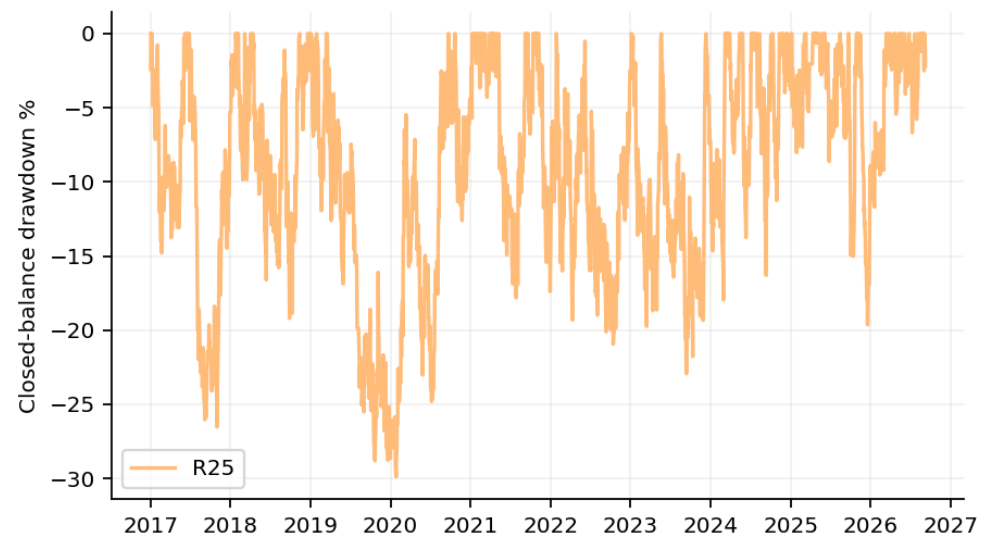
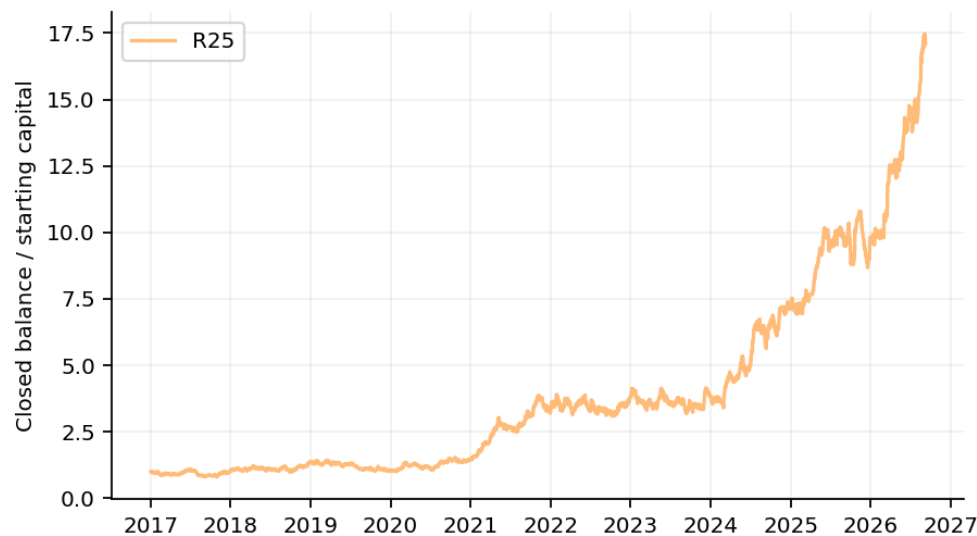
Rolling 3/6/12-month proxy worst returns: -23.33% / -23.58% / -25.75%; positive 12-month windows: 81.05%. Top 10 winners contribute 40.21% of net dollars and 25.97% of total R. With those top 10 R trades replaced by zero, replay CAGR is 22.68%.

R25 | ATR 0.65 | range only | risk 2.5%

Individual study 1/3 - performance, actual closed-balance curve and decision

The range cap modestly raises CAGR and Sharpe versus U25, but worsens equity drawdown and recovery duration. It does not meet the objective of materially reducing drawdown.

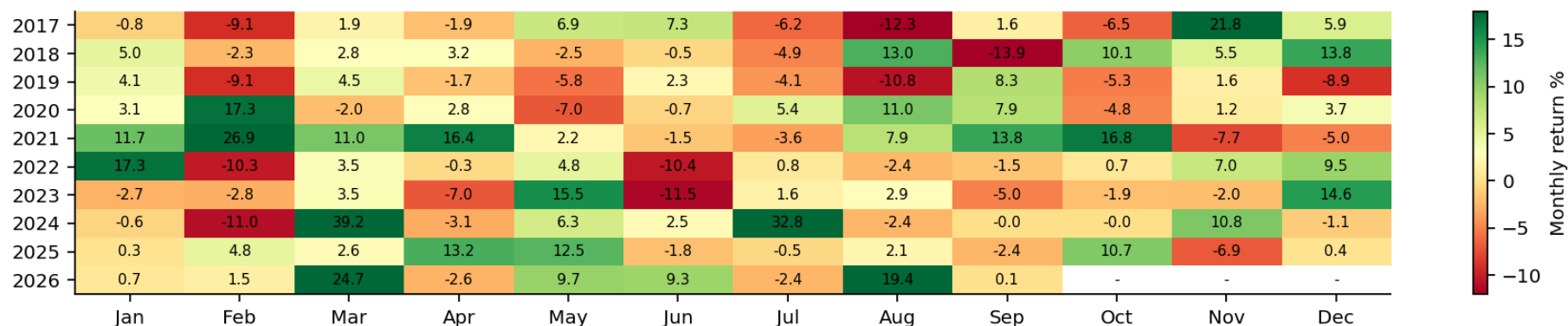
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
1,609,299.71	1,709,299.71	34.05	2,378	1.281	50.84
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
29.89	31.28	1.089	1.107	1.841	29.48



Average budget expectancy: 0.0550R/trade. Average winner: 0.637R; average loser: -0.547R. Trades/month across the full horizon: 20.5. Longest win/loss streaks: 13/10. Average holding time: 10.61 hours.

R25 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

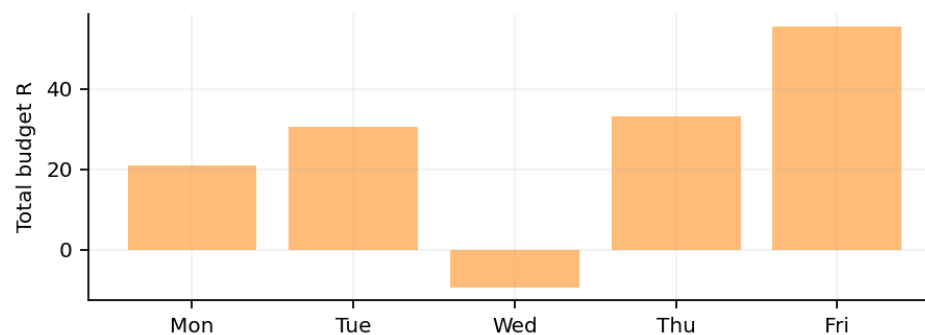
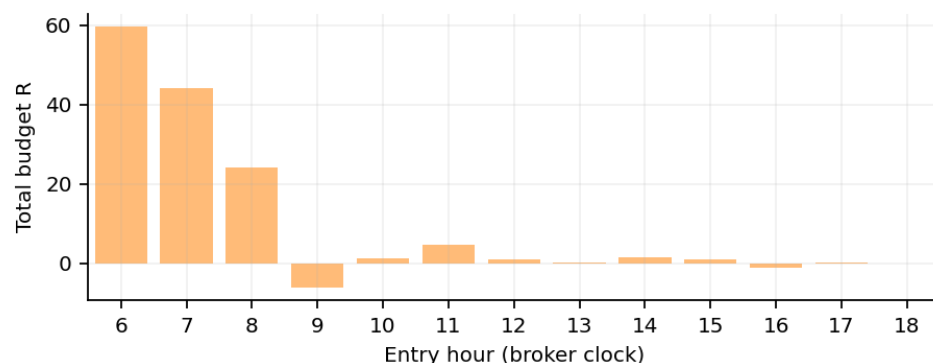


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	249	4,133	4.13	26.52	1.026	0.0126
2018	248	30,285	29.08	19.20	1.156	0.0491
2019	246	-32,201	-23.96	28.81	0.856	-0.0383
2020	245	42,985	42.05	20.47	1.206	0.0648
2021	253	179,437	123.58	17.82	1.416	0.1353
2022	251	54,037	16.65	20.94	1.079	0.0329
2023	251	6,616	1.75	22.94	1.010	0.0110
2024	245	325,537	84.49	16.30	1.393	0.1072
2025	229	272,618	38.35	19.63	1.204	0.0634
2026	161	725,854	73.81	6.68	1.743	0.1430

Positive months: 66/117; positive weeks: 291/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (39.24%); worst: 2018-09 (-13.88%). Best/worst weeks: 2023-12-04/2023-12-10 (17.69%) / 2018-09-10/2018-09-16 (-10.70%).

R25 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2020-01-27	2020-09-21	29.89	553
2017-06-27	2017-10-31	2018-01-24	26.52	211
2023-05-22	2023-09-13	2023-12-11	22.94	203

Side	Trades	Net profit \$	PF	Mean R
buy	1237	1,158,538	1.441	0.0758
sell	1141	450,761	1.145	0.0325

Actual initial SL exposure averaged 2.496% (range 2.467%-2.500%); worst closed trade -2.62%. Empirical daily VaR95/ES95: 2.50%/2.51%. 20-day block bootstrap balance-DD median / 95th percentile: 38.84% / 55.18%. These resampling levels are conditional stress estimates.

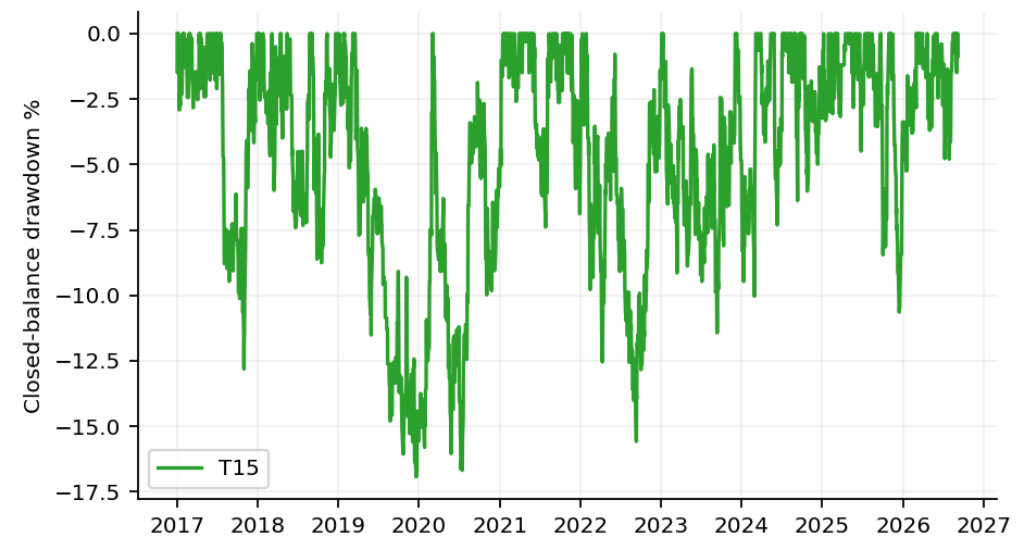
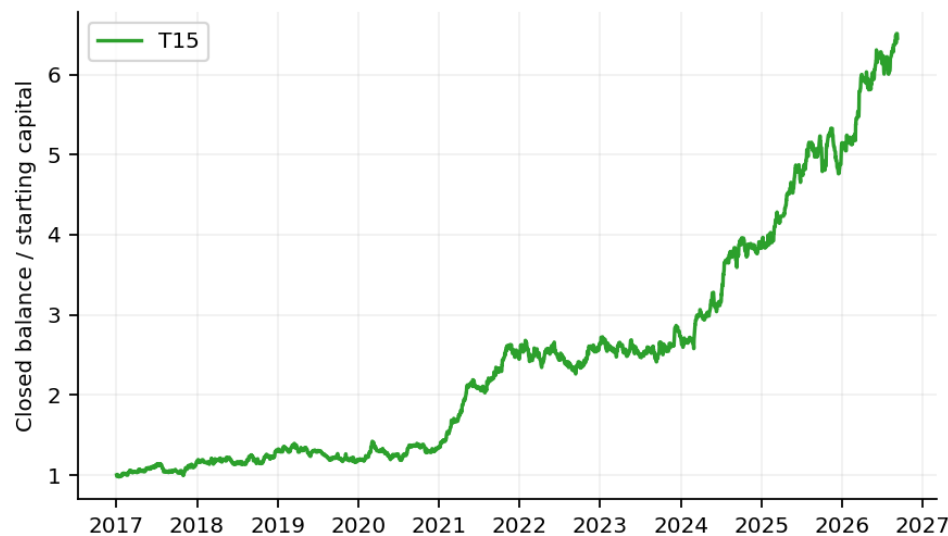
Rolling 3/6/12-month proxy worst returns: -23.47% / -23.04% / -25.93%; positive 12-month windows: 84.62%. Top 10 winners contribute 35.91% of net dollars and 25.31% of total R. With those top 10 R trades replaced by zero, replay CAGR is 23.49%.

T15 | ATR 0.65 | range + time | risk 1.5%

Individual study 1/3 - performance, actual closed-balance curve and decision

Preferred balanced growth replacement for L25. It improves CAGR, reported equity drawdown, Sharpe and recovery duration; retain L65 if the original lower absolute drawdown is required.

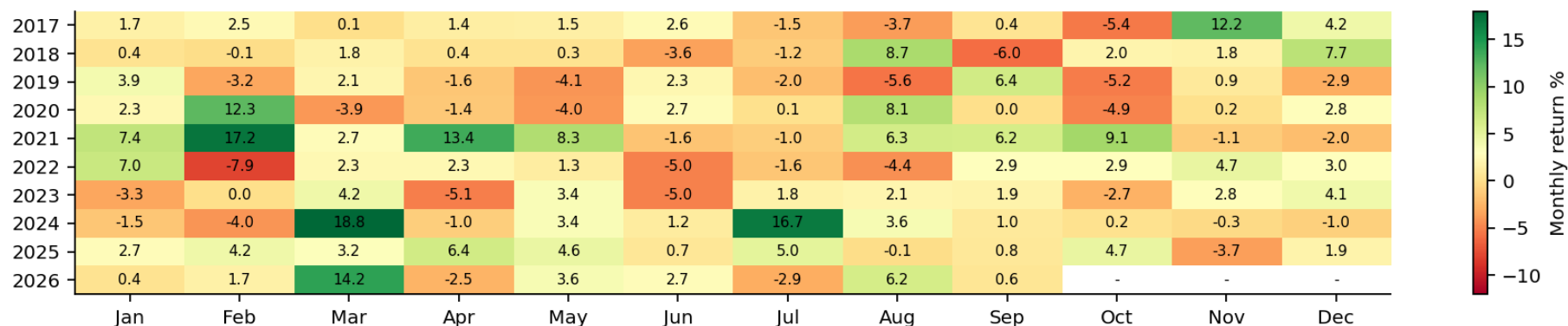
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
545,523.63	645,523.63	21.23	1,853	1.304	51.43
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
16.93	18.55	1.145	1.244	2.114	15.97



Average budget expectancy: 0.0717R/trade. Average winner: 0.662R; average loser: -0.553R. Trades/month across the full horizon: 15.9. Longest win/loss streaks: 9/9. Average holding time: 11.34 hours.

T15 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

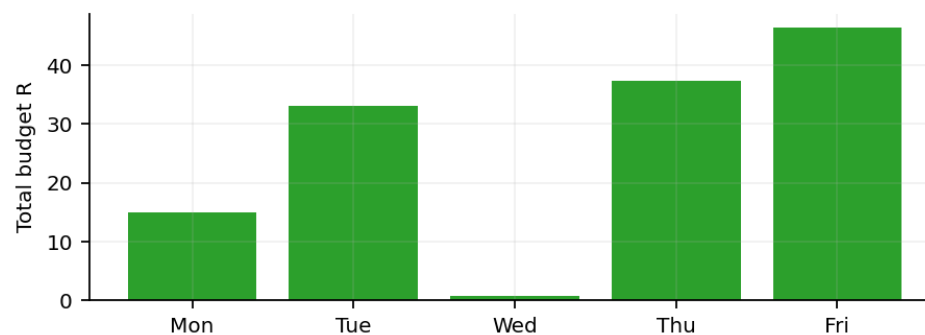
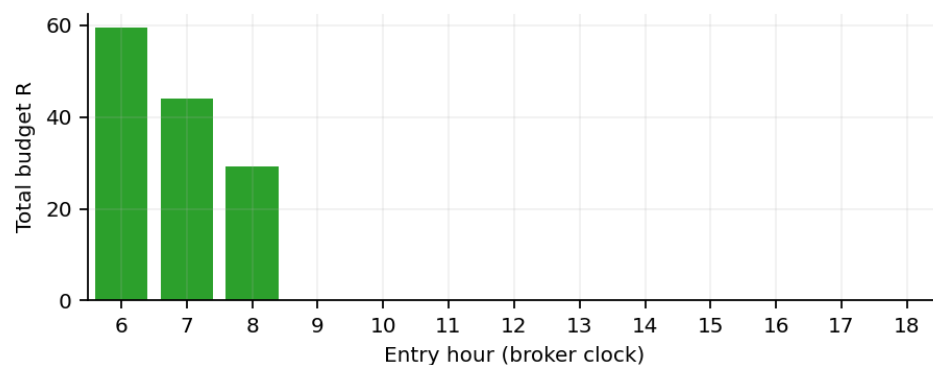


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	188	16,184	16.18	12.81	1.204	0.0571
2018	202	13,776	11.86	8.75	1.137	0.0413
2019	184	-12,247	-9.42	16.93	0.880	-0.0321
2020	185	16,136	13.71	16.68	1.154	0.0513
2021	195	112,946	84.38	7.39	1.837	0.2146
2022	200	16,578	6.72	15.58	1.068	0.0269
2023	197	9,415	3.57	11.43	1.041	0.0170
2024	196	110,582	40.54	7.31	1.458	0.1203
2025	185	131,512	34.30	10.64	1.405	0.1107
2026	121	130,641	25.37	4.79	1.567	0.1284

Positive months: 78/117; positive weeks: 281/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (18.80%); worst: 2022-02 (-7.93%). Best/worst weeks: 2024-03-04/2024-03-10 (8.98%) / 2025-09-29/2025-10-05 (-5.85%).

T15 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-12-20	2020-03-03	16.93	351
2020-03-05	2020-07-15	2021-01-18	16.68	319
2022-01-27	2022-09-12	2023-01-05	15.58	343

Side	Trades	Net profit \$	PF	Mean R
buy	978	355,902	1.411	0.0922
sell	875	189,621	1.204	0.0487

Actual initial SL exposure averaged 1.496% (range 1.465%-1.500%); worst closed trade -1.57%. Empirical daily VaR95/ES95: 1.50%/1.50%. 20-day block bootstrap balance-DD median / 95th percentile: 20.93% / 31.83%. These resampling levels are conditional stress estimates.

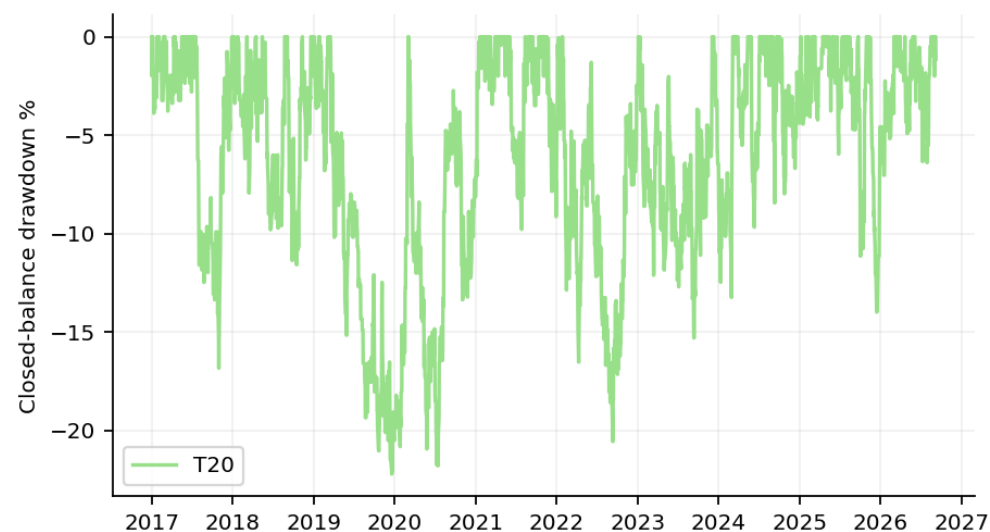
Rolling 3/6/12-month proxy worst returns: -13.07% / -13.32% / -12.10%; positive 12-month windows: 88.00%. Top 10 winners contribute 31.38% of net dollars and 24.23% of total R. With those top 10 R trades replaced by zero, replay CAGR is 15.47%.

T20 | ATR 0.65 | range + time | risk 2%

Individual study 1/3 - performance, actual closed-balance curve and decision

Conditional higher-growth alternative to T15. Higher CAGR comes with 24.11% equity drawdown; use instead of T15, not as another independent strategy.

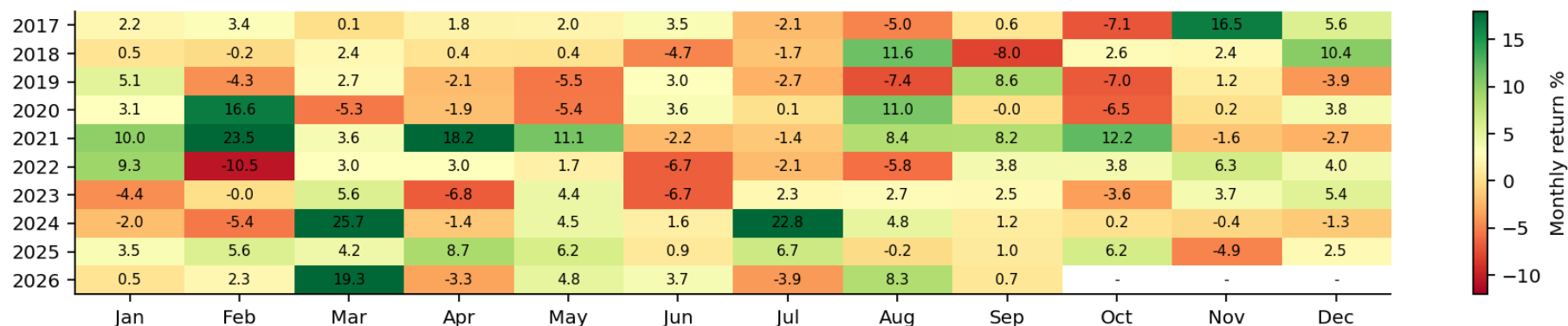
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
1,040,068.86	1,140,068.86	28.56	1,853	1.316	51.43
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
22.21	24.11	1.185	1.244	2.115	21.32



Average budget expectancy: 0.0718R/trade. Average winner: 0.663R; average loser: -0.554R. Trades/month across the full horizon: 15.9. Longest win/loss streaks: 9/9. Average holding time: 11.34 hours.

T20 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

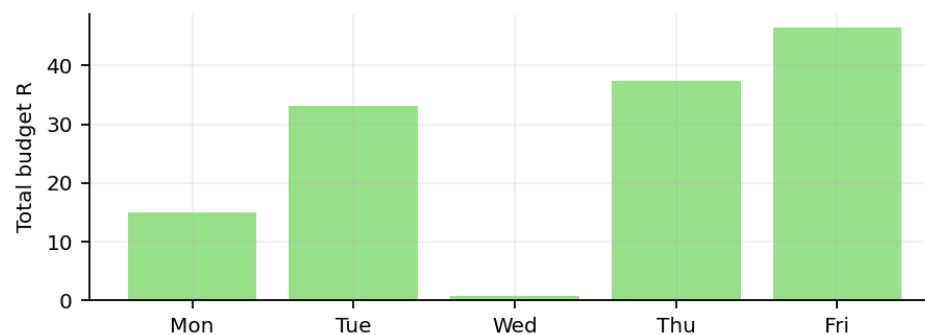
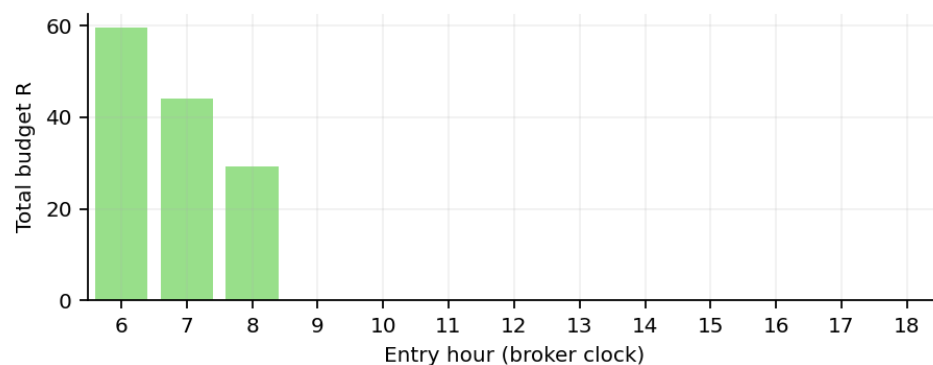


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	188	21,491	21.49	16.83	1.199	0.0570
2018	202	18,766	15.45	11.57	1.133	0.0413
2019	184	-17,964	-12.81	22.21	0.876	-0.0322
2020	185	22,026	18.01	21.79	1.147	0.0514
2021	195	180,175	124.85	9.78	1.788	0.2149
2022	200	26,931	8.30	20.56	1.062	0.0269
2023	197	14,360	4.09	15.29	1.035	0.0170
2024	196	206,677	56.50	9.67	1.445	0.1203
2025	185	271,948	47.51	13.98	1.392	0.1110
2026	121	295,659	35.01	6.39	1.561	0.1292

Positive months: 76/117; positive weeks: 281/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (25.70%); worst: 2022-02 (-10.48%). Best/worst weeks: 2024-03-04/2024-03-10 (12.11%) / 2025-09-29/2025-10-05 (-7.76%).

T20 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-12-20	2020-03-03	22.21	351
2020-03-05	2020-07-15	2021-01-20	21.79	321
2022-01-27	2022-09-12	2023-01-05	20.56	343

Side	Trades	Net profit \$	PF	Mean R
buy	978	669,865	1.424	0.0923
sell	875	370,204	1.216	0.0488

Actual initial SL exposure averaged 1.997% (range 1.977%-2.000%); worst closed trade -2.10%. Empirical daily VaR95/ES95: 2.00%/2.00%. 20-day block bootstrap balance-DD median / 95th percentile: 27.18% / 40.50%. These resampling levels are conditional stress estimates.

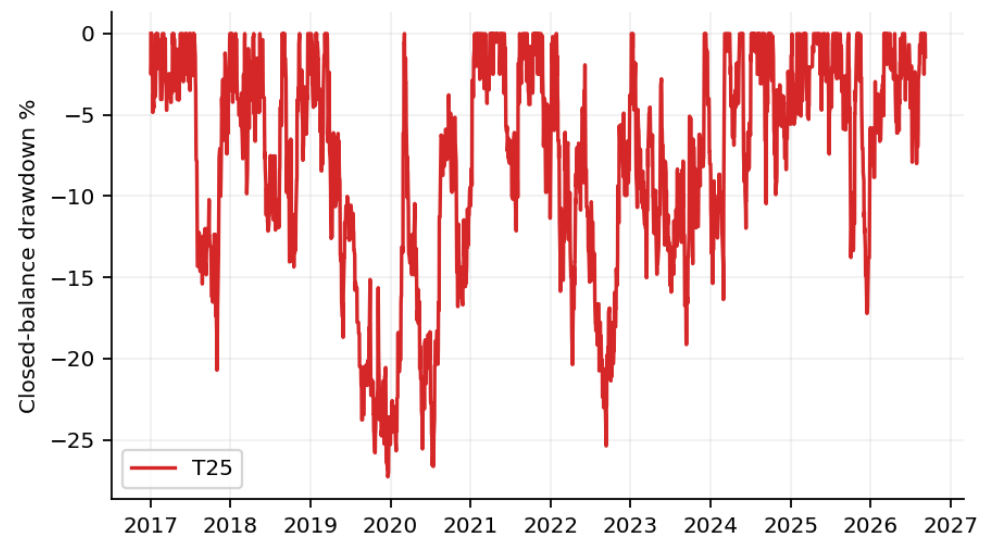
Rolling 3/6/12-month proxy worst returns: -17.18% / -17.50% / -16.17%; positive 12-month windows: 86.64%. Top 10 winners contribute 34.89% of net dollars and 24.22% of total R. With those top 10 R trades replaced by zero, replay CAGR is 20.54%.

T25 | ATR 0.65 | range + time | risk 2.5%

Individual study 1/3 - performance, actual closed-balance curve and decision

Highest CAGR and CAGR/equity-DD ratio in this batch, but 29.35% equity drawdown and substantially deeper bootstrap stress. Not the default for a drawdown-conscious allocation.

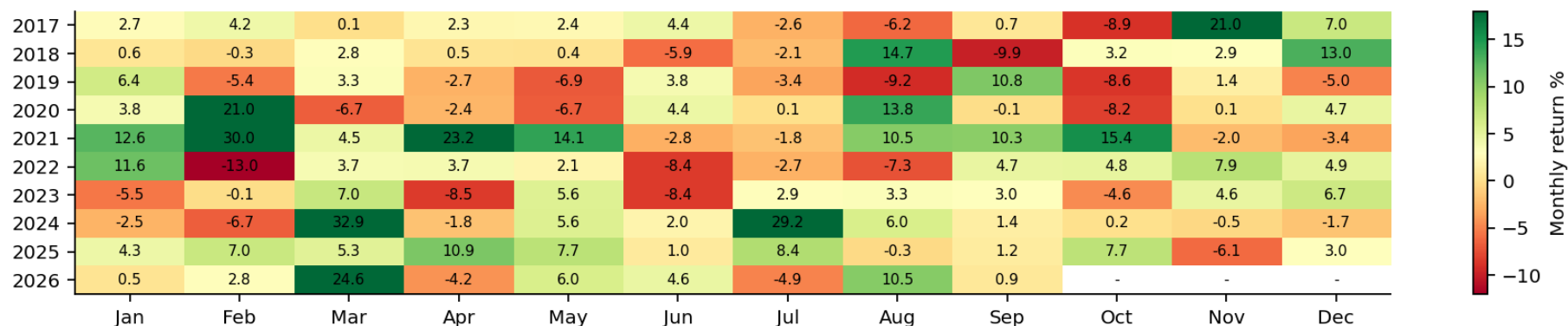
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
1,857,132.34	1,957,132.34	35.94	1,853	1.325	51.43
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
27.27	29.35	1.224	1.245	2.115	26.67



Average budget expectancy: 0.0718R/trade. Average winner: 0.663R; average loser: -0.554R. Trades/month across the full horizon: 15.9. Longest win/loss streaks: 9/9. Average holding time: 11.34 hours.

T25 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

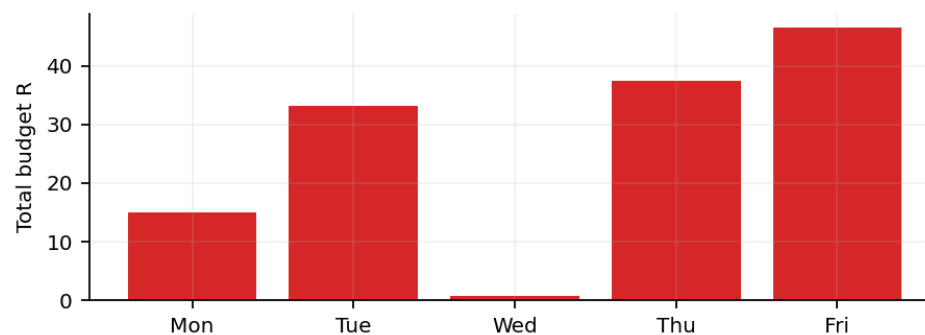
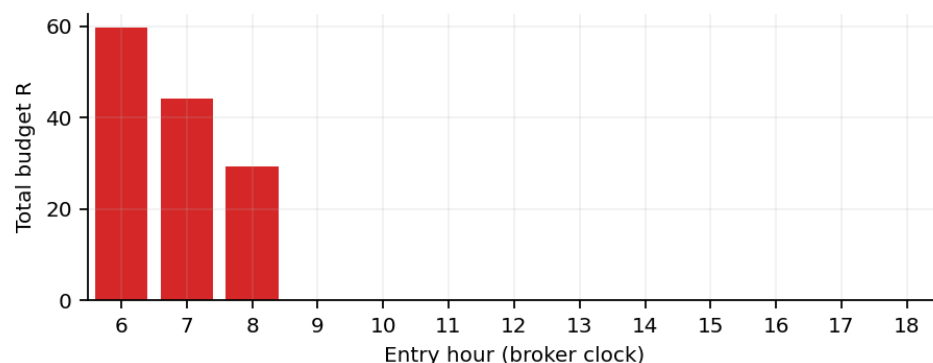


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	188	26,851	26.85	20.70	1.195	0.0571
2018	202	23,884	18.83	14.36	1.129	0.0414
2019	184	-24,462	-16.23	27.27	0.874	-0.0321
2020	185	27,812	22.03	26.63	1.139	0.0513
2021	195	266,599	173.02	12.13	1.741	0.2150
2022	200	40,096	9.53	25.37	1.057	0.0269
2023	197	19,703	4.28	19.13	1.030	0.0170
2024	196	354,876	73.86	11.97	1.434	0.1204
2025	185	514,161	61.55	17.21	1.379	0.1111
2026	121	607,613	45.02	7.99	1.553	0.1293

Positive months: 76/117; positive weeks: 281/506 calendar weeks in the test (including inactive weeks). Best month: 2024-03 (32.94%); worst: 2022-02 (-12.99%). Best/worst weeks: 2024-03-04/2024-03-10 (15.29%) / 2025-09-29/2025-10-05 (-9.64%).

T25 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-12-20	2020-03-05	27.27	353
2020-03-05	2020-07-15	2021-01-20	26.63	321
2022-01-27	2022-09-12	2023-01-05	25.37	343

Side	Trades	Net profit \$	PF	Mean R
buy	978	1,180,745	1.435	0.0923
sell	875	676,387	1.226	0.0489

Actual initial SL exposure averaged 2.497% (range 2.475%-2.500%); worst closed trade -2.62%. Empirical daily VaR95/ES95: 2.50%/2.50%. 20-day block bootstrap balance-DD median / 95th percentile: 33.13% / 48.23%. These resampling levels are conditional stress estimates.

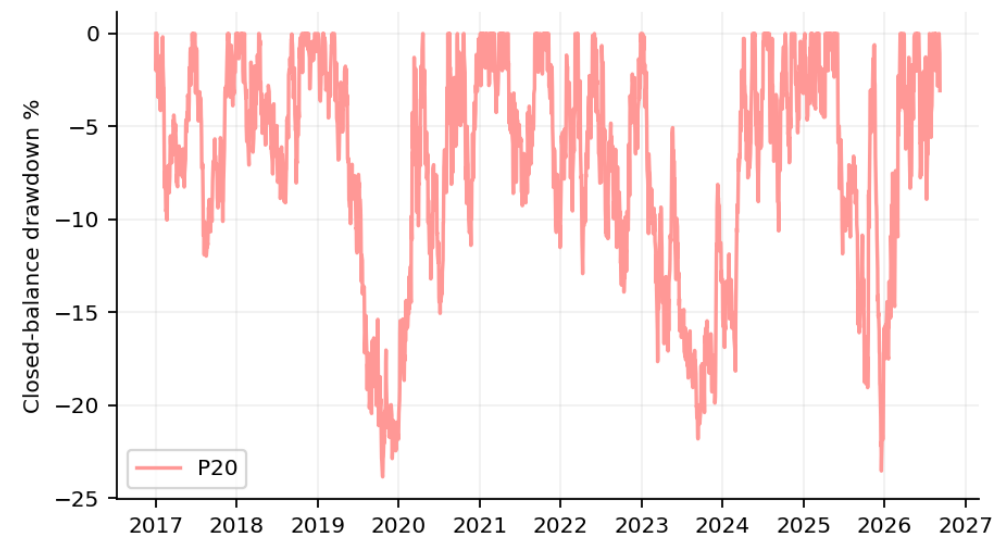
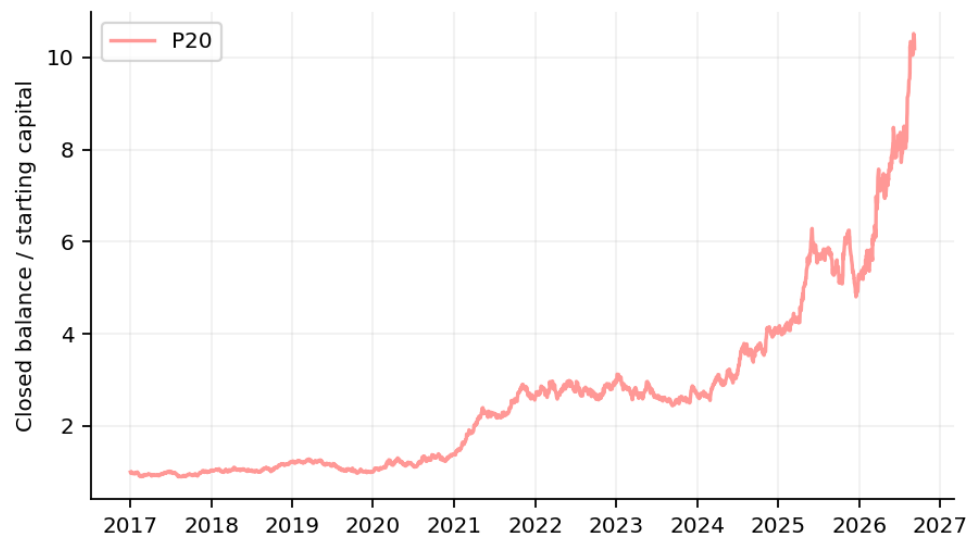
Rolling 3/6/12-month proxy worst returns: -21.12% / -21.59% / -20.23%; positive 12-month windows: 85.36%. Top 10 winners contribute 38.63% of net dollars and 24.23% of total R. With those top 10 R trades replaced by zero, replay CAGR is 25.48%.

P20 | Price SL 1% | unfiltered | risk 2%

Individual study 1/3 - performance, actual closed-balance curve and decision

Price-based stop delivers strong growth, but T20 has higher CAGR and lower equity drawdown. Keep as a research reference, not a preferred additional ORB sleeve.

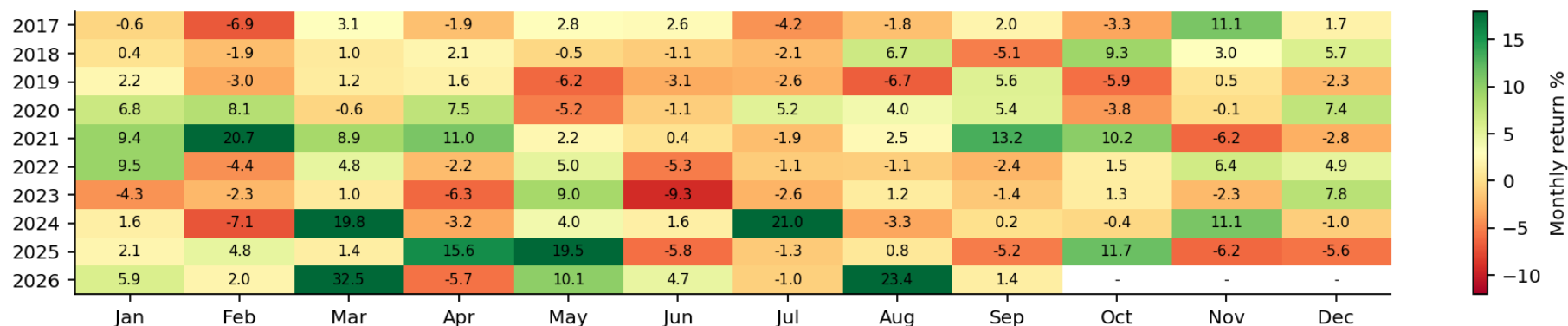
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
919,145.34	1,019,145.34	27.08	2,479	1.263	50.79
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
23.87	25.86	1.047	1.152	1.932	22.23



Average budget expectancy: 0.0518R/trade. Average winner: 0.574R; average loser: -0.487R. Trades/month across the full horizon: 21.3. Longest win/loss streaks: 14/11. Average holding time: 10.64 hours.

P20 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

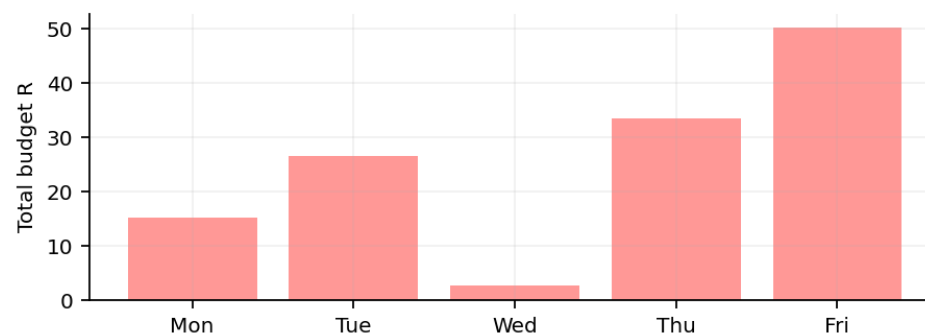
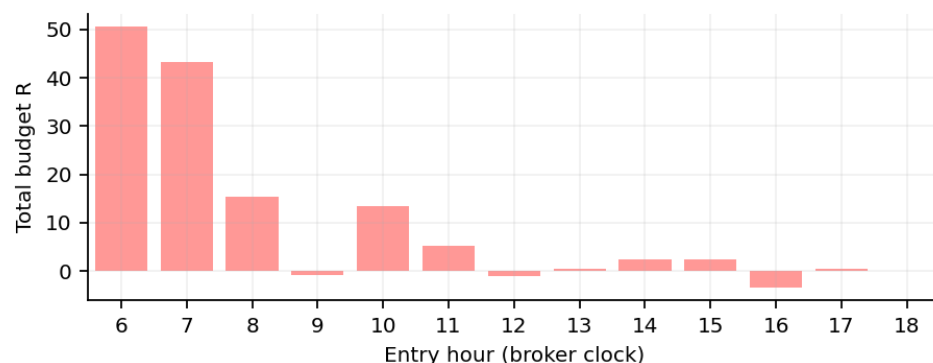


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	255	3,623	3.62	11.98	1.041	0.0092
2018	257	18,481	17.83	9.10	1.197	0.0345
2019	256	-21,830	-17.88	23.87	0.824	-0.0358
2020	257	37,860	37.76	15.06	1.208	0.0693
2021	257	120,335	87.11	11.51	1.517	0.1266
2022	257	39,876	15.43	13.91	1.100	0.0334
2023	256	-27,281	-9.14	21.82	0.923	-0.0144
2024	258	130,858	48.28	10.62	1.338	0.0808
2025	254	127,022	31.60	23.55	1.165	0.0607
2026	172	490,201	92.68	8.92	1.571	0.2028

Positive months: 66/117; positive weeks: 290/506 calendar weeks in the test (including inactive weeks). Best month: 2026-03 (32.46%); worst: 2023-06 (-9.26%). Best/worst weeks: 2026-08-03/2026-08-09 (12.16%) / 2026-06-08/2026-06-14 (-7.76%).

P20 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2019-10-22	2020-04-20	23.87	399
2025-06-03	2025-12-18	2026-03-13	23.55	283
2023-01-06	2023-09-13	2024-05-16	21.82	496

Side	Trades	Net profit \$	PF	Mean R
buy	1289	741,336	1.481	0.0744
sell	1190	177,809	1.091	0.0273

Actual initial SL exposure averaged 1.995% (range 1.984%-2.000%); worst closed trade -2.25%. Empirical daily VaR95/ES95: 2.00%/2.00%. 20-day block bootstrap balance-DD median / 95th percentile: 29.23% / 44.07%. These resampling levels are conditional stress estimates.

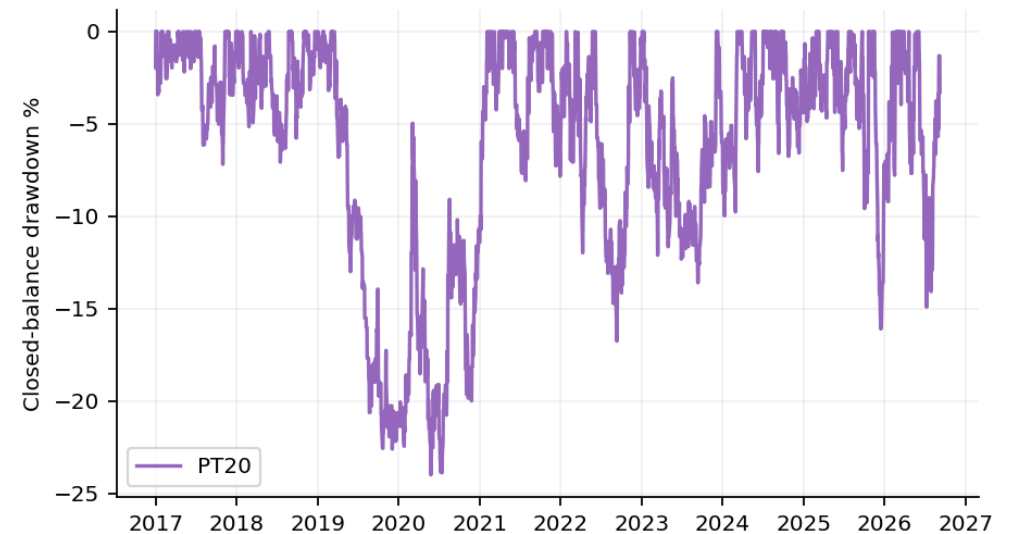
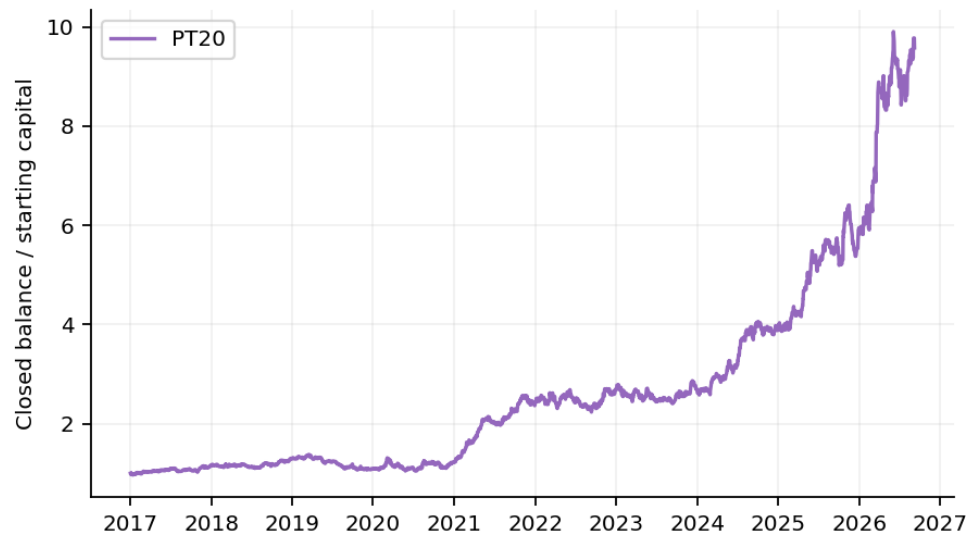
Rolling 3/6/12-month proxy worst returns: -15.71% / -21.29% / -18.78%; positive 12-month windows: 79.69%. Top 10 winners contribute 45.86% of net dollars and 27.30% of total R. With those top 10 R trades replaced by zero, replay CAGR is 18.51%.

PT20 | Price SL 1% | range + time | risk 2%

Individual study 1/3 - performance, actual closed-balance curve and decision

Highest daily Sharpe among these nine, but filters reduce CAGR versus P20 and recovery is slow. T20 beats its CAGR and equity drawdown; Sharpe alone does not settle selection.

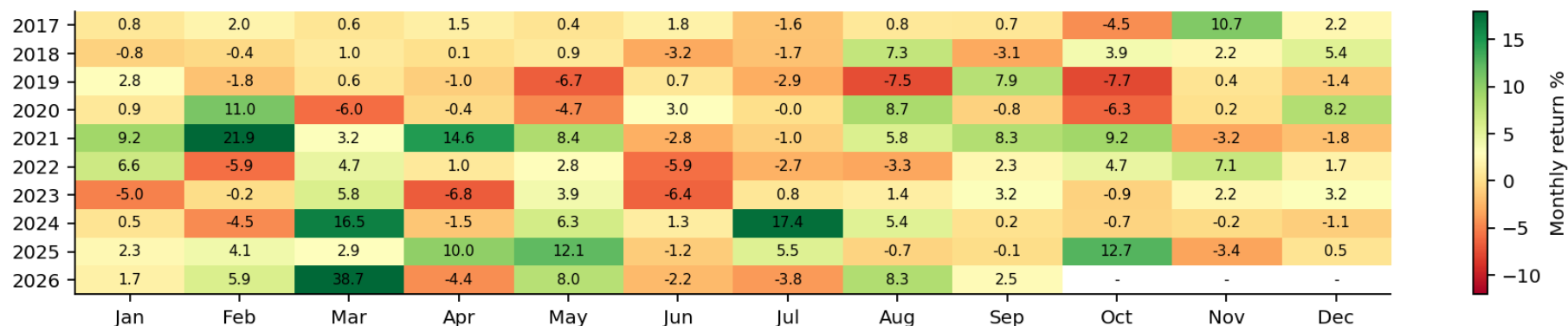
Net profit \$	Final balance \$	CAGR %	Trades	PF	Win %
858,015.19	958,015.19	26.27	1,853	1.336	51.59
Balance DD %	Equity DD %	CAGR / Eq DD	Sharpe	Sortino	Annual vol %
23.99	25.25	1.041	1.258	2.150	19.40



Average budget expectancy: 0.0660R/trade. Average winner: 0.584R; average loser: -0.486R. Trades/month across the full horizon: 15.9. Longest win/loss streaks: 10/8. Average holding time: 11.46 hours.

PT20 | Yearly and monthly consistency

Individual study 2/3 - complete monthly return heatmap; all cells in percent

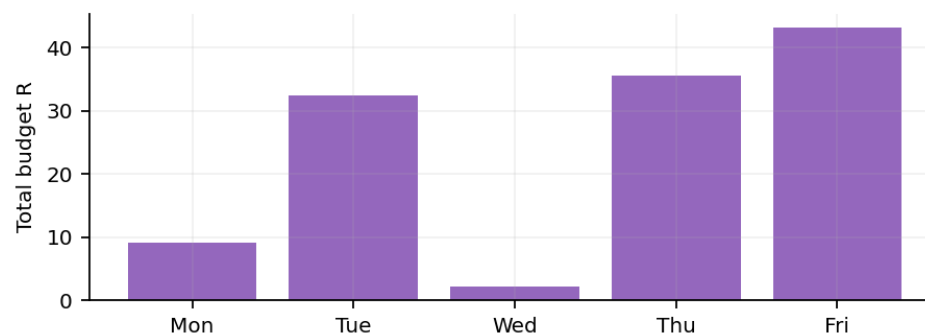
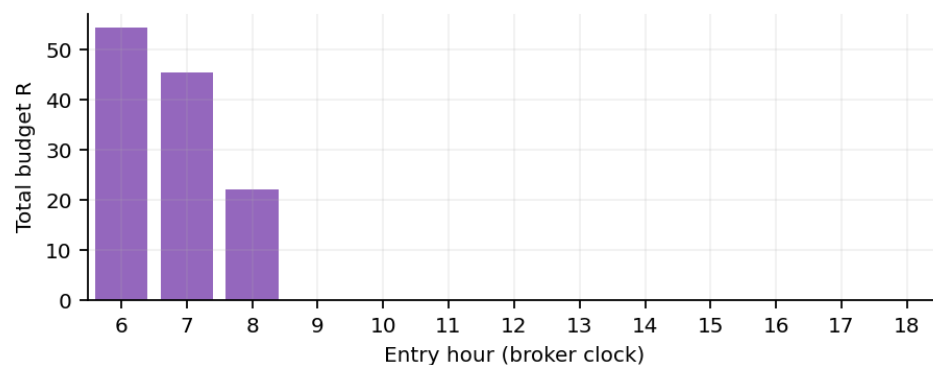


Year	Trades	Net profit \$	Return %	Within-year Bal DD %	PF	Mean R
2017	188	15,868	15.87	7.18	1.239	0.0414
2018	202	13,633	11.77	7.07	1.170	0.0298
2019	184	-21,054	-16.26	22.60	0.777	-0.0456
2020	185	13,671	12.61	20.03	1.101	0.0394
2021	195	116,143	95.11	8.07	1.765	0.1766
2022	200	30,147	12.65	16.75	1.107	0.0354
2023	197	949	0.35	13.60	1.004	0.0051
2024	196	118,931	44.15	7.57	1.422	0.0975
2025	185	206,474	53.18	16.09	1.408	0.1222
2026	121	363,255	61.08	14.91	1.512	0.2101

Positive months: 72/117; positive weeks: 287/506 calendar weeks in the test (including inactive weeks). Best month: 2026-03 (38.72%); worst: 2019-10 (-7.73%). Best/worst weeks: 2026-03-23/2026-03-29 (10.24%) / 2026-02-09/2026-02-15 (-7.30%).

PT20 | Behaviour, drawdowns and stress

Individual study 3/3 - timing diagnostics are descriptive, not new optimized filters



Peak	Trough	Recovery	Balance depth %	Duration days
2019-03-18	2020-05-26	2021-02-04	23.99	689
2022-06-07	2022-09-12	2022-11-11	16.75	157
2025-11-17	2025-12-15	2026-02-20	16.09	95

Side	Trades	Net profit \$	PF	Mean R
buy	978	520,179	1.439	0.0785
sell	875	337,836	1.247	0.0522

Actual initial SL exposure averaged 1.995% (range 1.983%-2.000%); worst closed trade -2.25%. Empirical daily VaR95/ES95: 2.00%/2.00%. 20-day block bootstrap balance-DD median / 95th percentile: 25.06% / 37.73%. These resampling levels are conditional stress estimates.

Rolling 3/6/12-month proxy worst returns: -16.83% / -19.30% / -18.96%; positive 12-month windows: 83.96%. Top 10 winners contribute 47.29% of net dollars and 28.31% of total R. With those top 10 R trades replaced by zero, replay CAGR is 17.85%.

Stress tests: historical DD is not a ceiling

Circular daily-return blocks preserve short local loss clustering; 2,000 resamples for each block length.

ID	Actual Bal DD %	Actual Eq DD %	5-day DD95 %	20-day DD95 %	20-day DD99 %	20-day CAGR5 %
U15	18.33	19.27	40.66	38.09	44.07	9.68
U20	24.00	25.16	50.79	47.90	54.87	12.29
U25	29.34	30.80	59.56	56.50	63.67	14.51
R25	29.89	31.28	58.04	55.18	62.14	15.57
T15	16.93	18.55	33.45	31.83	38.35	11.59
T20	22.21	24.11	42.29	40.50	48.00	15.10
T25	27.27	29.35	50.13	48.23	56.26	18.41
P20	23.87	25.86	43.56	44.07	50.07	13.31
PT20	23.99	25.25	37.66	37.73	45.10	13.37
L65	12.50	13.17	28.93	26.93	31.48	6.71
L25	20.66	21.05	33.06	31.76	36.72	8.60
P10	12.46	13.80	24.04	24.49	28.48	7.03

T15 has approximately 31.83% balance DD at the 95th percentile of the 20-day bootstrap, despite 18.55% reported historical equity DD. T25 rises to approximately 48.23% on the same bootstrap measure. Do not confuse either estimate with a guaranteed future bound or a calibrated 5% probability.

The bootstrap repeats observed returns; it cannot create missing market regimes, model execution failures, or unseen gaps. It is a fragility check, and it does not account for all the earlier variants examined during development.

Execution costs and concentration

Recorded commission/swap are zero. Additional per-lot costs below are sensitivity scenarios, not a claim about actual broker pricing.

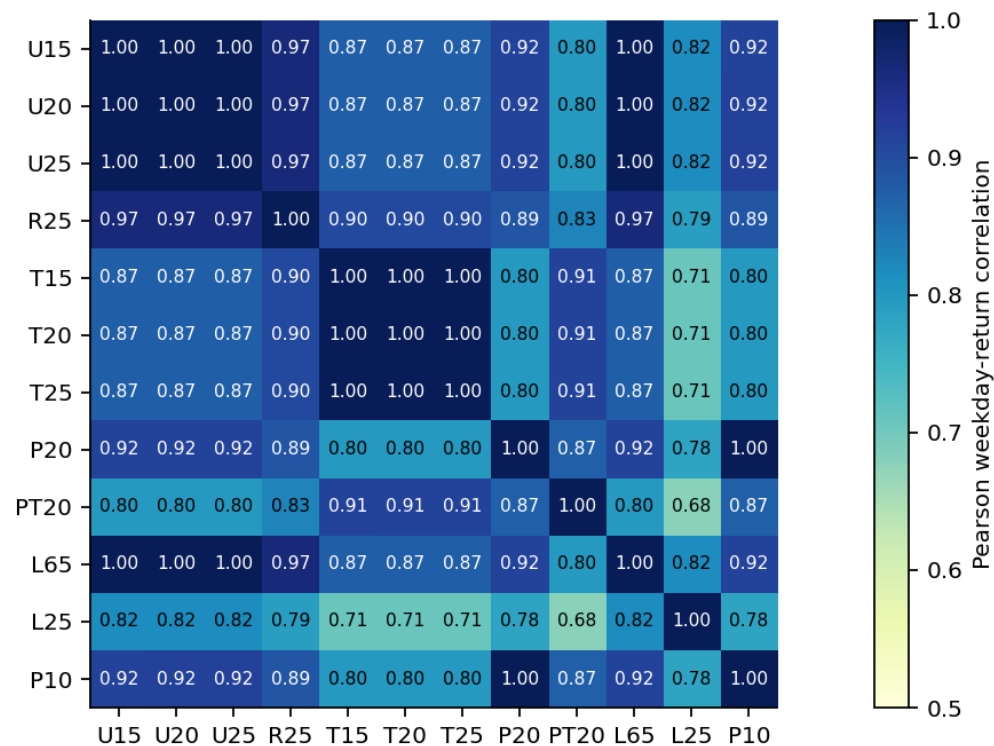
ID	Base PF	PF +\$7/lot	PF +\$20/lot	PF +\$27/lot	Mean R +\$27	Max recorded lots
U15	1.214	1.198	1.169	1.153	0.0328	4.12
U20	1.220	1.205	1.178	1.164	0.0329	6.87
U25	1.226	1.212	1.187	1.173	0.0329	12.97
R25	1.281	1.266	1.239	1.225	0.0353	11.19
T15	1.304	1.287	1.256	1.240	0.0519	4.47
T20	1.316	1.300	1.271	1.256	0.0519	7.73
T25	1.325	1.311	1.284	1.270	0.0520	14.92
P20	1.263	1.250	1.226	1.214	0.0368	4.80
PT20	1.336	1.322	1.297	1.284	0.0510	4.61
L65	1.206	1.188	1.157	1.141	0.0325	2.14
L25	1.220	1.199	1.161	1.141	0.0587	3.72
P10	1.240	1.224	1.197	1.182	0.0366	1.04

Costs are extra USD per round-trip lot deducted at the originally recorded volume. These scenarios do not rerun compounding or resize future trades. They should not be read as new CAGR estimates. Spread already embedded in fills should not be counted twice when calibrating real costs.

Higher risk and compounding can lead to large later position sizes. Historical order acceptance does not establish live capacity, margin headroom or liquidity. The observed average risk remains close to each input, so widespread minimum-lot or size-clipping distortion is not evident from these ledgers; broker limits still need an execution check.

Correlation: these remain one Gold ORB family

All weekday return series aligned; non-trading weekdays have zero return.



T15 versus T20/T25 is effectively perfectly correlated. T15-L65 correlation is 0.873; T15-L25 is 0.709. Different risk settings are not separate systems. A new portfolio component should be evaluated for behaviour during ORB drawdowns, not just a different filename.

Combining models: explicit capital allocation

Daily-rebalanced return simulation, total starting capital USD 100,000. Weights sum to one; no added leverage.

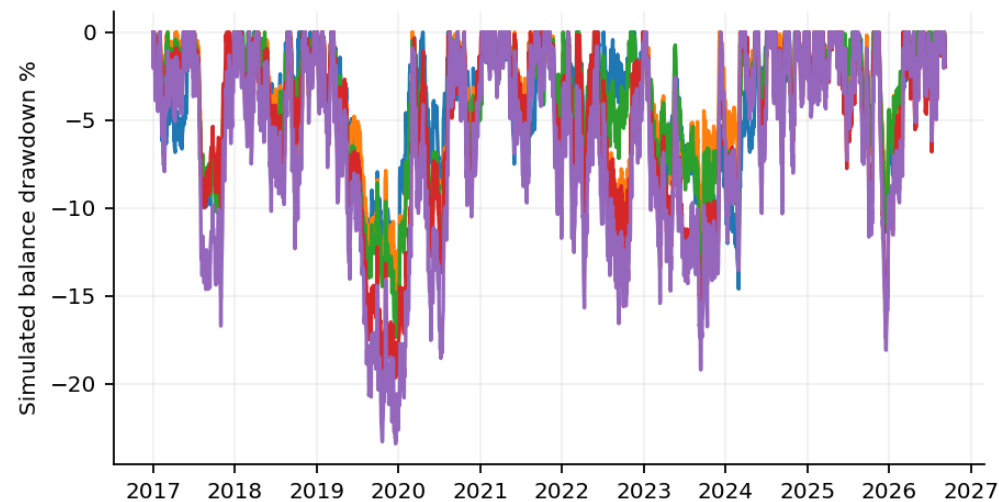
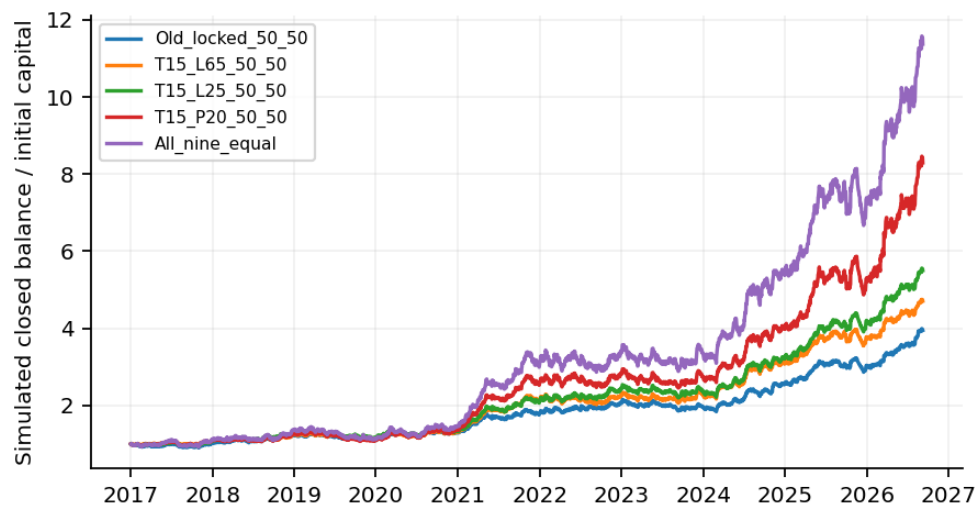
Mix	Allocation	CAGR %	Balance DD %	CAGR/Bal DD	Sharpe	20-day DD95 %
Old_locked_50_50	L65 50.0%, L25 50.0%	15.24	15.14	1.007	1.096	27.78
T15_L65_50_50	T15 50.0%, L65 50.0%	17.35	14.61	1.188	1.202	28.06
T15_L25_50_50	T15 50.0%, L25 50.0%	19.25	17.35	1.110	1.234	27.76
T15_P20_50_50	T15 50.0%, P20 50.0%	24.41	19.97	1.222	1.254	35.59
All_nine_equal	U15 11.1%, U20 11.1%, U25 11.1%, R25 11.1%, T15 11.1%, T20 11.1%, T25 11.1%, P20 11.1%, PT20 11.1%	28.53	23.40	1.219	1.206	42.52

A 50/50 capital mix of T15 and L65 implies approximately 0.75% + 0.50% risk against total capital when both take a trade. It does NOT mean applying full 1.5% and full 1% risk to the same account. Summing all nine full-risk EAs would be a much more leveraged, different experiment.

Each day's simulated return is the weighted sum of synchronized standalone daily returns, then compounded. This is a portfolio approximation, not a combined MT5 execution. No joint margin, order conflicts, intraday floating P&L, or same-symbol netting behaviour is simulated. Combined equity DD cannot be inferred from these exports.

Portfolio curves and rebalancing sensitivity

Simulated closed balances; not intratrade equity curves



Mix	Daily-rebalanced CAGR %	Daily-rebalanced DD %	No-rebalance CAGR %	No-rebalance DD %
Old_locked_50_50	15.24	15.14	15.25	15.31
T15_L65_50_50	17.35	14.61	17.87	14.78
T15_L25_50_50	19.25	17.35	19.22	17.46
T15_P20_50_50	24.41	19.97	24.45	19.80
All_nine_equal	28.53	23.40	29.07	23.65

No-rebalance results start each independent sleeve at its capital weight and let weights drift. The model uses proportionally scaled recorded balances; actual small-account lot rounding may differ. Neither approach makes the included strategies uncorrelated.

Portfolio yearly returns and interpretation

Daily-rebalanced combinations; no leverage beyond the defined capital weights

Year	Old_locked_50_50	T15_L65_50_50	T15_L25_50_50	T15_P20_50_50	All_nine_equal
2017	3.19	9.95	9.18	9.83	12.03
2018	19.45	12.95	18.54	14.94	21.09
2019	-11.06	-9.66	-10.69	-13.63	-17.30
2020	17.67	16.87	14.74	25.59	30.00
2021	37.41	61.47	57.09	85.98	107.57
2022	17.31	7.08	17.12	11.19	12.33
2023	-5.32	0.82	-2.57	-2.82	-1.10
2024	29.56	34.82	35.29	44.64	60.06
2025	19.23	22.64	30.75	33.37	37.23
2026	29.94	24.61	30.87	56.27	53.96

The all-nine equal-weight portfolio intentionally illustrates the result of combining every candidate. It repeats several near-identical ATR paths, so it overweights one family and is not the recommended five-system portfolio design.

Prefer one ORB configuration as the primary allocation. A deliberate L65/T15 blend can moderate exposure and retain some later entries, but any benefit must justify extra execution complexity. Use the remaining system slots for separately validated signals and instruments.

Methods, definitions and reproducibility

The source exports are authoritative for reported equity DD; all cash-based calculations are independently reconstructed.

$CAGR = (\text{ending balance} / \text{initial deposit})^{(365.25 / \text{elapsed calendar days})} - 1$. The horizon is 2017-01-01 to 2026-09-09. Year/month/week returns compound from opening to closing balance. Weekly periods are Monday-Sunday, clipped to test dates. September 2026 is partial.

Maximum relative balance DD includes initial deposit as a possible peak. Equity DD is copied from the explicitly labelled MT5 relative field. The percentage beside maximum currency drawdown can describe another episode. CAGR / equity DD and CAGR / balance DD are labelled separately; they are full-period Calmar-style ratios.

Budget R = net trade P/L / (entry balance x configured risk fraction). It normalizes position sizing but can differ from exact rounded SL risk. Actual initial SL exposure is verified using the USD 100-per-price-unit-per-lot conversion supported by every ledger. PF uses positive net P/L / absolute negative net P/L. Zero P/L is breakeven.

Daily returns use Mon-Fri closed balances with inactive days retained. Sharpe uses zero benchmark, sample daily standard deviation and $\sqrt{252}$. Sortino uses root mean square of negative daily returns including zero downside on other days. VaR/ES are empirical daily loss-tail statistics. Rolling 3/6/12-month proxies use 63/126/252 business days.

Bootstrap: circular blocks of 5 and 20 daily returns, 2,000 resamples, seed 19092026. Portfolio bootstrap uses synchronized portfolio returns, 20-day blocks, seed 230926. Synthetic paths do not carry independently observed equity DD. All comparisons are in-sample development diagnostics.

Primary definition reference: MetaTrader 5 Help, Testing Report: https://www.metatrader5.com/en/terminal/help/algotrading/testing_report (consulted for balance/equity relative drawdown definitions). Source evidence: nine newly attached XLSX files, three prior XLSX baselines, and the earlier Eight Models portfolio review identifying locked C and D.

Source traceability and complete supporting data

Every filename maps to the model register. The attached ZIP contains analysis scripts, input records and complete period/trade ledgers.

ID	Source filename	SHA-256 prefix
U15	UF_65_15RISK.xlsx	59993d9b33e108cf79f84547
U20	UF_65_2RISK.xlsx	64cf643625e97ce4fe3a0025
U25	UF_65_25RISK.xlsx	fac39d21b8311e11b6e9696e
R25	UF_65_RANGE_25_RISK.xlsx	e7f95720a8df906916e8425a
T15	UF_65_RANGE_TIME_15RISK.xlsx	387b757f8a5f34fd6269b306
T20	UF_65_RANGE_TIME_2RISK.xlsx	7bbefddeed57d0bc5533f2b3
T25	UF_65_RANGE_TIME_25RISK.xlsx	45f047153d475581cfb1c701
P20	1PERCENTSL_2RISK.xlsx	9e152ae418476e11ad476019
PT20	1PERCENTSL_WITHRANGETIME_2RISK.xlsx	8ac9185de7ce2a6b430f85f0
L65	UF_65_1_RISK.xlsx	8f4b5a71022f95ea10e31600
L25	UF_25_06_RISK.xlsx	944b9fd36a7571af362357eb
P10	UF_25_PRICE_SL1PERCENT.xlsx	9ec8077e9ac702bc9c68eec7

The PDF attachment ORB_Risk_Research_Data.zip contains: analysis.json (source header and every input, summary, hourly/weekday/side metrics, periods, drawdowns, costs, stress); extra.json (filters, replays and allocations); every reconstructed trade/daily ledger; all monthly/weekly/yearly CSVs; removed-trade lists; portfolio series; and the scripts used for analysis/report generation.

A desktop PDF reader with attachment support may be needed to extract the ZIP. Exact input values are retained there so that a filename or stale inactive input cannot silently change the interpretation. Files are read-only sources; no source workbook was modified.

Final section | Are these better than the locked models?

The decision separates historical efficiency, absolute drawdown and portfolio purpose.

Candidate	CAGR %	Eq DD %	CAGR / Eq DD	Longest balance DD days	Decision
L65	13.40	13.17	1.018	432	Keep for lowest-risk role
L25	16.86	21.05	0.801	552	Superseded as default growth candidate
T15	21.23	18.55	1.145	351	Preferred new balanced growth
T20	28.56	24.11	1.185	351	Optional higher exposure
T25	35.94	29.35	1.224	353	Aggressive; not default

Versus locked L25: yes, T15 is the stronger historical choice. CAGR rises by 4.37 percentage points, equity DD falls by 2.50 points, and CAGR/equity-DD improves from 0.801 to 1.145. Sharpe rises from 1.036 to 1.244. This is not merely a higher-return/higher-drawdown trade.

Versus locked L65: more efficient growth, but not lower absolute drawdown. T15 raises CAGR from 13.40% to 21.23%, while equity DD rises from 13.17% to 18.55%. If the lower loss depth was the reason for locking L65, it remains valid. T15 should not automatically replace that conservative role.

The same-drawdown replay supports the filtered logic, but its 1% implementation is still untested in this batch. The new evidence changes the growth selection more decisively than it changes the conservative selection.

Final selection | What to take forward

Research shortlist, not a statement of live or prop-firm readiness

Objective	Choice	Justification
One ORB with balanced growth	T15: ATR 0.65, risk 1.5%, range 0.00-0.60, entry 06:05-08:55	Best fit to the requested return/DD trade-off. 21.23% CAGR, 18.55% equity DD, stronger consistency than the old growth choice.
Preserve original conservative role	L65: ATR 0.65, risk 1%, both filters OFF	13.17% equity DD remains lower than every new run. Keep until a filtered low-risk MT5 rerun provides an actual replacement comparison.
Deliberately accept more drawdown	T20 instead of T15	28.56% CAGR with 24.11% equity DD. It is the same filtered signal at higher exposure, not an extra diversified system.
Do not default to	T25, U25, R25 or all nine at full account risk	A better ratio does not offset a roughly 29%-31% historical equity drawdown for a drawdown-conscious mandate.

My selection: replace L25 with T15 on the research shortlist; retain L65 for the conservative role. If selecting only one growth-oriented ORB, choose T15. If the absolute drawdown ceiling is near the old 13%, keep L65 and test the filtered model at 1% before changing it.

Before deployment, freeze the chosen rules, confirm the actual tester mode and broker costs, and validate forward or on genuinely untouched data. Retest the shortlist on a common executable and audit a small set of entries, stops and exits. No further minute-by-minute optimization is justified by this report. For a funded account, assess the exact account-specific loss rules separately; these full-period DD figures do not establish compliance.

The improvement is real within this historical sample: the ATR 0.65 filters produce a better trade selection than the unfiltered variants. Future stability remains to be demonstrated. Keep the next system-development effort focused on a different source of returns rather than another risk setting of this same ORB.